BMJ Open

BMJ Open is committed to open peer review. As part of this commitment we make the peer review history of every article we publish publicly available.

When an article is published we post the peer reviewers' comments and the authors' responses online. We also post the versions of the paper that were used during peer review. These are the versions that the peer review comments apply to.

The versions of the paper that follow are the versions that were submitted during the peer review process. They are not the versions of record or the final published versions. They should not be cited or distributed as the published version of this manuscript.

BMJ Open is an open access journal and the full, final, typeset and author-corrected version of record of the manuscript is available on our site with no access controls, subscription charges or pay-per-view fees (http://bmjopen.bmj.com).

If you have any questions on BMJ Open's open peer review process please email info.bmjopen@bmj.com

BMJ Open

Hypertensive Disorders in Pregnancy and Risk of Asthma in Offspring: A Systematic Review and Meta-Analysis

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-046769
Article Type:	Original research
Date Submitted by the Author:	09-Nov-2020
Complete List of Authors:	Li, Ping; West China Second University Hospital, Sichuan University, Department of Pediatrics; Key Laboratory of Birth Defects and Related Diseases of Women and Children (Sichuan University) Ministry of Education, China Xiong, tao; West China Second University Hospital, Sichuan University, Department of Pediatrics; Key Laboratory of Birth Defects and Related Diseases of Women and Children (Sichuan University) Ministry of Education, China, Hu, Yong; West China Second University Hospital, Sichuan University, Department of Pediatrics; Key Laboratory of Birth Defects and Related Diseases of Women and Children (Sichuan University) Ministry of Education, China
Keywords:	Hypertension < CARDIOLOGY, Asthma < THORACIC MEDICINE, PAEDIATRICS, Community child health < PAEDIATRICS, Fetal medicine < OBSTETRICS, PERINATOLOGY

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

For peer review only

Hypertensive Disorders in Pregnancy and Risk of Asthma in Offspring: A Systematic Review and Meta-Analysis

Ping Li, MD,[1,2] Tao Xiong*, MD PhD [1,2,3], Yong Hu, MD [1,2]

[1] Department of Pediatrics, West China Second University Hospital, Sichuan University, Chengdu, China

[2] Key Laboratory of Birth Defects and Related Diseases of Women and Children (Sichuan University) Ministry of Education, China

[3] Deep Underground Space Medical Center, West China Hospital, Sichuan University, Chengdu, China

***Correspondence**

Dr. Tao Xiong

No. 20, Section Three, South Renmin Road, Chengdu, China

Tel: 86 13730652940

Email: tao_xiong@126.com

Word count: 2917

For peer review only

ABSTRACT

Objectives Hypertensive disorders in pregnancy (HDP) has a controversial association with an increased risk of asthma in offspring. No systematic review of this topic has been performed. The aim of this systematic review was to summarize the available evidence regarding the association between HDP and the risk of asthma in offspring.

Design Systematic review and meta-analysis.

Methods On the basis of a prepared protocol, a systematic search of PubMed, EMBASE, the Cochrane Library and Web of Science was performed using a detailed search strategy from database inception through Jan 17, 2020. English-language cohort, case-control and cross-sectional studies that reported diagnoses of maternal HDP and asthma in offspring were included. Meta-analysis of Observational Studies in Epidemiology (MOOSE) guidelines were followed throughout. The estimated pooled odds ratios (ORs) for HDP and asthma in offspring were calculated from studies and meta-analyzed using random-effects models.

Results Ten cohort studies, with a total of 252,450 cases and 6,270,430 participants, were included. According to the Newcastle-Ottawa Scale (NOS), the overall methodological quality was good since 8 studies were of high quality and 2 studies were of moderate quality. After

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

controlling for potential confounders, HDP increased the risk of asthma in offspring, with a pooled adjusted odds ratio (aOR) of 1.19 (95% CI, 1.12- 1.26). Subgroup analyses according to HDP subgroups, sibling design, study quality, study location, offspring ages, singleton status, exposure assessment, outcome assessment, and adjusted factors showed similar results.

Conclusions Exposure to HDP may be associated with an increased risk of asthma in offspring. Improved pediatric surveillance is warranted for infants exposed to maternal HDP; surveillance may be beneficial for early asthma diagnosis and intervention.

PROSPERO registration number CRD42020148250.

Keywords: hypertensive disorders in pregnancy, preeclampsia, blood pressure, asthma, offspring, childhood, meta-analysis

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Strengths and limitations of this study

- Studies finally included in this meta-analysis were prospective cohort design, and out of ten were of high quality, which could eliminate various confounding factors.
- An extremely large sample size including more than six million participants was included, thus our findings are potential more robust than those of any individual study.
- The study was based on a pre-prepared protocol and followed the Meta-Analyses guidelines and the Meta-analysis Of Observational Studies in Epidemiology guidelines.
- Hypertensive disorders in pregnancy and asthma diagnostic criteria varied among the included studies, which may increase clinical heterogeneity.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

INTRODUCTION

Hypertensive disorders in pregnancy (HDP), which are estimated to affect 5% to 15% of all pregnancies, are recognized as some of the most common obstetrical complications.[1 2] There are four subtypes of HDP, including preeclampsia-eclampsia, chronic hypertension, chronic hypertension with superimposed preeclampsia, and gestational hypertension (GH).[3] HDP subtypes are serious threats to the health of offspring. Recent studies have confirmed that HDP are closely associated with a risk of offspring diseases, including high blood pressure,[4] congenital heart defects,[5] obesity,[6] autism spectrum disorder, attention-deficit/hyperactivity disorder, low cognitive function, anxiety /depression, and other neurodevelopmental disorders.[7 8]

Theoretically, HDP induce a detrimental in utero environment, causing systemic inflammation and oxidative stress in the vulnerable fetal lung. Therefore, the development of the fetal lung may be disturbed by abnormal inflammation and immune function. These disturbances may persist through late life in offspring, leading to subsequent respiratory diseases. HDP may have great impacts on the fetal airway structure and respiratory and immune system development during the prenatal period, significantly increasing the

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

susceptibility of offspring to asthma.[9 10] Asthma, which affects 1-18% of the population, is the most common lung disease associated with abnormal inflammation and immune function.[11 12] It is characterized by variable symptoms of wheezing, shortness of breath, chest tightness and/or cough, and variable expiratory airflow limitation.[13-15] Since asthma can cause recurrent school absences, emergency department visits and hospitalizations, it has been regarded as a major public health challenge worldwide.

The etiology of asthma is not completely clear. In addition to genetic and environmental factors,[16 17] it is widely accepted that asthma originates early in life.[18] Negative events in the perinatal period increase the risk of asthma and impair lung function later in life.[19 20] Asthma symptoms beginning in adulthood may have originated in childhood.[21] Recurrent wheezing and other asthma-like symptoms usually begin as early as the first few weeks or months after birth. This highlights the need for early recognition and interventions for risk factors of asthma to minimize the development of asthma and its potential long-term sequelae. Recent epidemiological studies have reported that maternal HDP may be a potential risk factor for asthma in offspring.[22-24] However, evidence of the causal association between maternal HDP and asthma in offspring is inconsistent.[25-27] Hence, clarifying this association is extremely

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

important to elucidate the pathogenesis of asthma and facilitates early recognition of and interventions for asthma in high-risk populations. Therefore, we performed the current systematic review and meta-analysis to synthesize the available evidence regarding the relationship between HDP and asthma in offspring.

METHODS

Data sources and search strategy

The systematic review included studies that met the following inclusion criteria: a target population of pregnant women and their offspring; exposure to HDP; a comparison group with no HDP; and a primary outcome of asthma, with secondary outcomes consisting of recurrent wheezing/wheezing, lung/pulmonary function, active asthma, asthma exacerbations or asthma treatment. The protocol for this systematic review and meta-analysis was registered in the international prospective register of systematic reviews (PROSPERO, ID: CRD42020148250) and published in BMJ OPEN.[28]

We followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines[29] and Meta-analysis of Observational Studies in Epidemiology (MOOSE)

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

guidelines.[30] Two reviewers (P.L. and T.X.) independently conducted a systematic search of the literature in PubMed, EMBASE, the Cochrane Library and Web of Science from database inception through Jan. 17, 2020. Search terms relating to HDP and asthma were combined according to the principles of Boolean logic (using AND, OR, or NOT). The full search strategy is included in eTable 1 in the Supplement. In addition, the reference lists of the included studies were manually searched to further identify eligible studies. The authors of 3 studies were contacted for additional information. Searches were limited to human studies published in English. No restrictions were placed on publication date, location of the study, or the age of participants.

Study selection

Two investigators (P.L. and Y.H.) independently inspected the titles and abstracts of the citations and obtained the full texts. When consensus on eligibility could not be achieved, a third reviewer (T.X.) was involved in the discussion. Eligibility criteria for inclusion in the systematic review included all the following: 1) cohort, case-control, or cross-sectional studies in which a diagnosis of HDP was reported and asthma was the outcome of interest; 2) studies

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

in which the association between HDP and asthma was part of the main objective of the study; 3) studies that contained original data and confirmed HDP through medical records and/or physician-diagnosed self-reporting; and 4) literature that was peer-reviewed. The exclusion criteria were as follows: 1) conference abstracts, letters to the editor, reviews and commentary articles; 2) studies with overlapping data; and 3) studies without raw data.

Data extraction

Two reviewers (P.L. and Y.H.) independently extracted data from the eligible studies using a standardized data extraction form. The titles and abstracts of studies retrieved from each database were stored and managed in the EndNote reference manager. For each included study, we extracted the following information: first author's last name, year of publication, study location, study design, sample size, exposure (HDP and its subgroups), outcome diagnostic criteria, offspring age at diagnosis, adjusted/matched confounding variables and effect estimates [risk ratios (RRs) or odds ratios (ORs)] with 95% confidence intervals (CIs). Any discrepancies were resolved by consensus with a third reviewer (T.X.).

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Bias and quality assessment

Quality assessments of the included studies were conducted by 2 reviewers (P.L. and Y.H.) independently, and any discrepancies were resolved by a third reviewer (T.X). The quality of cohort studies was assessed using the Newcastle-Ottawa Scale (NOS).[31] Study quality was classified into 3 categories: high quality (scored 7-9), moderate quality (scored 4-6) and low quality (scored 0-3).

Statistical analysis

Data were analyzed using Stata version 14.0 (StataCorp, College Station, TX, USA). The ORs with 95% CIs were used as a measure of the association between HDP and the risk of asthma in offspring among studies. The hazard ratios, incidence rates and RRs were directly considered ORs in the pooled analysis. We included adjusted ORs and used a random-effects model to calculate the pooled ORs. The I^2 statistic (significance level, >50%) was applied to examine heterogeneity. Forest plots were constructed to show the study-specific OR estimates and the pooled OR estimate. Subgroup analyses were stratified by HDP subgroups, sibling design, study quality, study location, offspring ages, singleton status, exposure

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

assessment, outcome assessment, and adjusted factors. Sensitivity analysis was performed by omitting one study by turns to evaluate the changes in the pooled OR. The potential risk of publication bias was estimated by inspection of a funnel plot. Publication bias was further assessed by Egger's test and Begg's test (significance level, $P < 0.05$), as well as the "trim-and-fill" procedure.

RESULUTS

Literature search and selection

The original search produced 1,247 unique results after the removal of duplicates. Of these, twenty-nine full-text articles were reviewed after screening the titles and abstracts. Of those, nineteen articles were excluded because they did not meet the inclusion criteria. Finally, ten studies[22-27 32-35] were included in this study. Reasons for exclusion are outlined in Figure 1.

Study characteristics

The characteristics of the 10 included studies are shown in Table 1. The studies were published between 2003 and 2019. All the selected studies[22-27 32-35] were cohort studies, and

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Table 1. Characteristics of the selected studies included in the systematic review and meta-analysis.

Author (year)	Study design	Country	Data source	Sample size	Ascertainment of HDP	Ascertainment of Asthma	Offspring age at diagnosis	Study Quality
Mirzakhani et al (2019) [24]	cohort	United States	VDAART	67 (mothers with preeclampsia) 739 (mothers without preeclampsia) 28 (mothers with asthma and preeclampsia) 294 (mothers with asthma without preeclampsia)	medical records (ACOG)	maternal-reported (physician-diagnosed)	3yrs	M
Wilmink et al (2018) [26]	cohort	Netherlands	Generation R Study	206 (mothers with GH) 91 (mothers with preeclampsia or HELLP syndrome) 4,783 (control)	medical records (ISSHP，ACOG)	maternal-reported (physician-diagnosed)	10yrs (8.6~12yrs)	H
Stokholm et al (2017) [23]	cohort	Denmark	Registry-based cohort	62,728 (mothers with preeclampsia) 1,635,910 (mothers without preeclampsia)	medical records (ICD-8, ICD-10)	medical records (ICD-10)	0~15yrs	H
			$COPSAC_{2000}$ Birth Cohort	20 (mothers with asthma and preeclampsia) 317 (mothers with asthma without preeclampsia)	medical records (ICD-10)	medical records (pediatrician- diagnosed)	7yrs	
Shaheen et al (2016) [27]	cohort	UK	ALSPAC	161 (mothers with preeclampsia) 1,140 (mothers with GH) 308 (mothers with HbP) 6,486 (normotensive)	medical records (ISSHP criteria14)	maternal-reported (doctor-diagnosed)	7yrs	H
Magnus et al (2016) [31]	cohort	Norway	Registry-based study	16,329 (mothers with preeclampsia) 390,578 (control)	medical records	medical records	7yrs	H
Liu et al (2015) [22]	cohort	Denmark	DMBR	nested case-control study: 30,767 (mothers with preeclampsia) 1,239,965 (mothers without preeclampsia) case-sibling study: 3,608 (mothers with preeclampsia) 143,704 (mothers without preeclampsia)	medical records (ICD-8, ICD-10)	medical records (ICD-10)	3~7yrs	H
Byberg et al (2014) [25]	cohort	Norway	Stavanger study	FU1 (F 10.8yrs, M 11.8yrs): 214 (mothers with preeclampsia) 366 (mothers without preeclampsia) FU2 (F, M: 12.8yrs): 169 (mothers with preeclampsia) 279 (mothers without preeclampsia)	medical records	FU1:maternal-reported (physician-diagnosed) FU2: self-reported (ISAAC)	10~12yrs	H
Aspberg et al (2010) [30]	cohort	Switzerland	Registry-based study	41,389 (mothers with preeclampsia) 1,296,083 (control)	medical records	medical records	2~15yrs	M
Rusconi et al (2007) [29]	cohort	Italy	SIDRIA	862 (mothers with hypertension or preeclampsia) 14,729 (control)	medical records	maternal-reported	6~7yrs	H
Nafstad et al (2003) [28]	cohort	Norway	Registry-based study	98,188 (mothers with preeclampsia) 1,428,341 (control)	medical records (ICD-8)	derived from NIAR (doctor-diagnosed,ICD-7, ICD-9)	7.5~25.5yrs	H

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Abbreviations: HDP, hypertension disorder in pregnancy; GH, gestational hypertension; HELLP, hemolysis, elevated liver enzymes, and low platelets; HbP, hypertension before pregnancy; VDRRT, Vitamin D Antenatal Asthma Reduction Trial; ISSHP, International Society for the Study of Hypertension in Pregnancy; ACOG, American College of Obstetricians and Gynaecologists; $COPSAC_{2000}$, Copenhagen Prospective Studies on Asthma in $Childhood_{2000;}$ ALSPAC, Avon Longitudinal Study of Parents and Children; DMBR, Danish Medical Birth Registry; ISAAC, International Study of Asthma and Allergies in Childhood; SIDRIA, Italian Studies of Respiratory Disorders in Childhood and the Environment; MBRN, Medical Birth Registry of Norway; NIAR, National Insurance Administration Register.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

there was a total of 252,450 cases and 6,270,430 participants. The sample sizes of the studies ranged from 580 to 1,698,638, and 4 studies[22 23 32 34] had sample sizes of more than 1 million. Nine studies[22 23 25-27 32-35] were performed in Europe, and one study[24] was performed in the United States. Four studies[27 32 33 35] included children aged 6 to 7 years old, and two[25 26] included children aged 10 to 12 years old. In 2 cohort studies,[22 35] the authors used sibling analysis in which children with the same mother were included, to control for shared genetic and social confounders. According to the NOS, the overall methodological quality was good since 8 studies[22 23 25-27 32 33 35] were of high quality and 2 studies[24 34] were of moderate quality (Supplemental eTable 2). The confounders that were adjusted for in the studies are shown in eTable 3 in the Supplement.

Results of the meta-analyses

Primary outcome: HDP and risk of asthma in offspring

Ten studies[22-27 32-35] in which a diagnosis of HDP was reported and asthma was the outcome of interest were identified. Seven studies[23-27 32 35] described the incidence of HDP ranged from 2.0% to 14.0% (mean, 6.7%; median, 5.6%; interquartile range [IQR], 3.3%-10.0%), and an

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

average of 4.6% mothers with preeclampsia (IQR, 2.3%-6.9%). The incidence of asthma ranged from 4.9% to 25.5% (mean, 14.7%; median, 12.5%; IQR, 6.5%-23.4%) in offspring of mothers diagnosed with HDP and 3.6%-17.5% (mean, 8.9%; median, 8.6%; IQR, 4.8%-13.3%) in the controls.[22-27 33-35] Fourteen estimates from 10 unique studies included adjusted estimates and were included in the meta-analysis. Figure 2 displays the adjusted estimates, producing a pooled adjusted odds ratio (aOR) of 1.19 (95% CI, 1.12- 1.26), suggesting an increased risk for asthma among offspring of mothers with HDP. Statistically significant heterogeneity was found across studies (I^2 = 72.5%).

Secondary outcomes

There is only a few data for secondary outcomes including recurrent wheezing/wheezing, active asthma, asthma exacerbations or asthma treatment in included studies. Only two studies reported physiological parameters of lung function. One study reported a decreased forced expiratory volume in one second (FEV_1), and another study reported a reduced forced expiratory flow between 25% and 75% (FEF_{25-75}) in offspring of mothers with preeclampsia.[25 27]

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Subgroup and sensitivity analyses

We performed subgroup analyses to explore heterogeneity according to subgroups of HDP, study quality, study location, offspring ages, singleton births, exposure assessment, outcome assessment, and adjusted factors. The results of the subgroup analyses are outlined in Table 2. The subgroup analysis examining the associations between subtypes of HDP and asthma in offspring resulted in aORs of 1.18 (95% CI, 1.11-1.25) for preeclampsia and 1.26 (95% CI, 0.98-1.61) for other HDP (Supplemental eFigure 1). The sibling analysis failed to reveal a relationship between preeclampsia and asthma in offspring, with a pooled aOR of 1.06 (95% CI, 0.99-1.14). Studies with high quality yielded significant aOR of 1.17 (95% CI, 1.09-1.25). In addition, significant aORs were observed in studies conducted in both Europe and the United States. For offspring ages, aORs in three age groups were significant, but not for offspring aged 10-12 years old. The aOR in the subgroup of singletons was slightly higher than that in not singletons, with a decreased heterogeneity of I^2 = 47.2%. The aORs were all significant for any type of exposure assessment (HDP from self-reports or medical records) and for any type of outcome assessment (asthma from maternal-reports or medical records). For the subgroup analyses by adjusted factors, the aORs were generally significant by adjustment for maternal

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Table 2. Pooled ORs and heterogeneity of the subgroup analyses.

Variable	No. of Studies (No. of estimates)	I^2 (%)	OR (95%CI)	Heterogeneity Statistic	*P* for Heterogeneity
Category of HDP					
Preeclampsia	9 (10)	77.0	1.18 (1.11-1.25)	39.11	<0.001
Other	3 (4)	62.5	1.26 (0.98-1.61)	8.01	0.046
Sibling design (only preeclampsia)					
Yes	2 (2)	0.0	1.06 (0.99-1.14)	0.01	0.935
No	7 (8)	75.8	1.15 (1.05-1.26)	28.92	<0.001
Study quality					
Moderate	2 (2)	21.6	1.28 (1.08-1.51)	1.28	0.259
High	8 (12)	66.9	1.17 (1.09-1.25)	33.26	<0.001
Location (only preeclampsia)					
Europe	8 (9)	78.5	1.17 (1.10-1.25)	37.15	<0.001
United States	1 (1)	0.0	1.62 (1.02-2.57)	NA	NA
Offspring ages (years)					
6-7	4 (6)	55.6	1.23 (1.10-1.38)	11.26	0.046
10-12	2 (4)	0.0	0.90 (0.61-1.33)	0.29	0.962
3	1 (1)	NA	1.62 (1.02-2.57)	NA	NA
Others [a]	3 (3)	92.5	1.17 (1.08-1.27)	26.55	<0.001
Only singletons					
Yes	6 (7)	47.2	1.24 (1.18-1.31)	11.35	0.078
No	4 (7)	0.0	1.10 (1.06-1.13)	3.07	0.799
Exposure assessment					
Self-report	1 (1)	NA	1.71 (1.25-2.35)	NA	NA
Medical records	9 (13)	70.1	1.18 (1.11-1.25)	40.16	<0.001
Outcome assessment					
Maternal-report	4 (7)	42.0	1.26 (1.05-1.52)	10.34	0.111
Medical records	6 (7)	82.8	1.18 (1.11-1.26)	34.92	<0.001
Maternal smoking during pregnancy					
adjusted	7 (11)	30.7	1.23 (1.17-1.28)	14.43	0.154
not adjusted	3 (3)	29.9	1.10 (1.03-1.17)	2.85	0.240
Maternal asthma					
adjusted	5 (8)	50.4	1.22 (1.12-1.33)	14.10	0.049
not adjusted	5 (6)	84.5	1.17 (1.05-1.29)	32.16	<0.001

[a] Others include 0 ~ 25.5 years old.

Abbreviations: OR, odds ratio; CI, confidence intervals; HDP, hypertension disorder in pregnancy; NA, not applicable

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

smoking during pregnancy (aOR, 1.23; 95% CI, 1.17-1.28) and maternal asthma (aOR, 1.22; 95% CI, 1.12-1.33). Meanwhile, the heterogeneity obviously decreased by adjusted for maternal smoking during pregnancy (I^2 = 30.7%) and maternal asthma (I^2 = 50.4%). The sensitivity analysis, which was performed by omitting each study one at a time, showed that the overall pooled adjusted ORs did not vary substantially (Supplemental eTable 4).

Publication bias

Visual inspection of the funnel plot revealed potential publication bias (Supplemental eFigure 2), although both Egger's test and Begg's test were not statistically significant (Egger's test, *P* = 0.709; 95% CI, -1.26-1.80; Begg's test, *z* = 0.770 (continuity corrected), *Pr* > |*z*| = 0.443 (continuity corrected)). Therefore, we further applied the trim-and-fill method to adjust for funnel plot asymmetry,[36] but the results showed that no trimming was performed, and the data were unchanged (''no trimming performed, data unchanged''). There was no indication of publication bias with Duval's trim-and-fill method (no new studies added).

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

DISCUSSION

To the best of our knowledge, this is the first systematic review and meta-analysis to assess the relationship between maternal HDP and asthma in offspring. The results are robust because they are based on 10 observational studies with moderate-high quality and a large sample of 6,270,430 participants. The estimates in original studies were carefully adjusted for potential confounders. We found that maternal HDP exposure was associated with approximately 20% increased odds of asthma in offspring compared with nonexposure. The results of the subgroup analysis examining the association between preeclampsia and asthma in offspring provided a significant aOR of 1.18,[22-27 32 34 35] and the association between other HDP and asthma in offspring resulted in a marginally significant aOR of 1.26.[26 27 33]

In the meta-analysis, 80% of the selected studies were identified to be of high quality,[22 23 25-27 32 33 35] according to the NOS for cohort studies. Five studies included birth registry-based cohorts.[22 23 32 34 35] The NOS scores of the included cohort studies ranged from 6 to 9, with a mean of 7.5, indicating high quality according to the NOS. Our meta-analysis has provided strong evidence to show a significantly increased risk of asthma in offspring exposed to maternal HDP. However, the worldwide generalizability of the results should be interpreted

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

cautiously because of the geographic restriction of the included studies, which mostly comprise participants from Europe or the United States. Future clinical investigations performed in other regions are much needed.

High heterogeneity was detected in this meta-analysis. Therefore, we performed subgroup analyses to explore the cause of the heterogeneity. Notably, we found that the heterogeneity decreased significantly when studies that considered only singletons and maternal smoking during pregnancy as a confounding factor were included, revealing that the number of offspring and maternal smoking during pregnancy may have contributed to the heterogeneity. Another explanation for heterogeneity may be due to variability in asthma incidence in offspring among included studies. The incidence of asthma ranged from 4.9% to 25.5% in offspring of mothers diagnosed with HDP and 3.6%-17.5% in the controls.

The genetic susceptibility of offspring to asthma was evaluated through three analyses in this study. First, since the family situation and genetic factors are similar in siblings, we performed a subgroup analysis of two-sibling-design studies,[22 35] and found that there was a borderline significant relationship between preeclampsia and asthma in offspring (aOR, 1.06; 95% CI, 0.99-1.14). Second, a positive relationship (aOR, 1.22; 95% CI, 1.12-1.33) was

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

observed between HDP and asthma in offspring when adjusted for maternal asthma.[22 24 25 27 35] The discrepancy among the two analyses may be due to the limited sample size or other unadjusted factors. A fundamental role of genetics should be noted in future studies evaluating maternal HDP and asthma in offspring.

Strengths and limitations

This systematic review has several strengths. It had an extremely large sample size including more than six million participants. All the ten studies were prospective cohort designs, and eight out of ten were of high quality. In addition, our study was based on a pre-prepared protocol and followed the PRISMA guidelines and MOOSE guidelines.[29 30 37] However, there are several potential limitations in this review. First, the included studies were limited to the English language, and potential non-English-language studies may have been missed. Second, the different assessments of HDP and asthma used among the studies may potentially introduce misclassification bias. HDP and asthma diagnostic criteria varied among the included studies, which may increase clinical heterogeneity.[38] Several studies[22 32 34 35] followed the diagnosis of asthma per the National Insurance Administration Register or National Prescribed Drug Register, which might exclude offspring with mild asthma. Third,

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

important confounding factors were not fully adjusted for in the included studies. Therefore, although an apparent association exists between HDP and asthma in offspring, future research is needed to identify a comprehensive set of confounders to assess whether this association is causal or attributable to residual or unmeasured confounding. Although there was a significant influence of HDP on asthma in offspring, the effects of HDP on physiological parameters of lung function, asthma exacerbations, or asthma treatment need further consideration in future studies.

Exposure to maternal HDP in early life may increase the risk of developing asthma later in life. This finding provides important evidence of the long-term consequences in offspring exposed to maternal HDP. Furthermore, it supports the fetal origin hypothesis for the development of asthma. Early surveillance and interventions are warranted for high-risk children of mothers with HDP.

CONCLUSION

Our study found a significantly increased risk of asthma in offspring of mothers with HDP; this result was derived from a large population and mostly high-quality studies. Additional robust

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

research is needed to address key limitations in the literature. We advocate for improved pediatric surveillance for infants exposed to maternal HDP, as increased surveillance may be beneficial for the early diagnosis of and intervention for asthma.

Acknowledgments

We acknowledge suggestions from Imti Choonara (Emeritus Professor in Child Health and Editor in Chief of BMJ Paeds Open, University of Nottingham, the Medical School, Derby, UK) on the conception and design of the analysis as well as other valuable advice regarding this manuscript.

Contributors

The study was conceived by P.L. and T.X. P.L., T.X., and Y.H. developed the eligibility criteria, search strategy, methodological quality assessment, data extraction and data summary plan. P.L. and T.X. wrote the protocol. T.X. supervised the work. All authors approved the final version.

Funding

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

This work is supported by the National Science Foundation of China (No. 81300525 to Tao Xiong, No. 81701540 to Ping Li); the Science and Technology Bureau of Sichuan Province (2020YJ0298, 2020YFS0109); Health Commission of Sichuan Province (20PJ071); Deep Underground Space Medical Center (DUGM201809); and the Fundamental Research Funds for the Central Universities (SCU2019C4005).

Competing Interests

None declared.

Patient consent for publication

Not required.

Provenance and peer review

Not commissioned; externally peer reviewed.

Data sharing statement

No data sharing available.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

REFERENCES

1. Gillon TE, Pels A, Dadelszen PV, et al. Hypertensive disorders of pregnancy: A systematic review of international clinical practice guidelines. *Plos One* 2014;9:122.

2. Olson-Chen C, Seligman NS. Hypertensive Emergencies in Pregnancy. *Critical Care Clinics* 2016;32:29-41.

3. Obstetriciansgynecologists ACO. Hypertension in pregnancy. Report of the American College of Obstetricians and Gynecologists' Task Force on Hypertension in Pregnancy. *Obstetrics & Gynecology* 2013;122:1122-31.

4. Pinheiro T, Brunetto S, Ramos J, et al. Hypertensive disorders during pregnancy and health outcomes in the offspring: a systematic review. *Journal of developmental origins of health and disease* 2016;7:391-407.

5. Boyd HA, Basit S, Behrens I, et al. Association Between Fetal Congenital Heart Defects and Maternal Risk of Hypertensive Disorders of Pre gnancy in the Same Pregnancy and Across Pregnancies. *Circulation* 2017;136:39-48.

6. Zheng JS, Liu H, Ong KK, et al. Maternal Blood Pressure Rise During Pregnancy and Offspring Obesity Risk at 4 to 7 Years Old: The Jia xing Birth Cohort. *J Clin Endocrinol Metab*

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

2017;102:4315-22.

7. Tuovinen S, Eriksson JG, Kajantie E, et al. Maternal hypertensive pregnancy disorders and cognitive functioning of the offspring: a systematic re view. *J Am Soc Hypertens* 2014;8:832-47.

8. Maher GM, O'Keeffe GW, Kearney PM, et al. Association of Hypertensive Disorders of Pregnancy With Risk of Neurodevelopmental Disorders in Offsp ring: A Systematic Review and Meta-analysis. *JAMA Psychiatry* 2018;75:809-19.

9. Harding R, Maritz G. Maternal and fetal origins of lung disease in adulthood. *Semin Fetal Neonatal Med* 2012;17:67-72.

10. Postma DS, Bush A, van den Berge M. Risk factors and early origins of chronic obstructive pulmonary disease. *Lancet* 2015;385:899-909.

11. Boulet LP, Reddel H, Bateman E, et al. The Global Initiative for Asthma (GINA): 25 years later. *Eur Respir J* 2019; 54:1900598, doi:10.1183/13993003.00598-2019.

12. Huang K, Yang T, Xu J, et al. Prevalence, risk factors, and management of asthma in China: a national cross-sectional study. *Lancet* 2019; 394: 407-18.

13. Lebold KM, Jacoby DB, Drake MG. Inflammatory mechanisms linking maternal and

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

childhood asthma. *J Leukoc Biol* 2020; 108:113-21.

14. Holgate ST. Innate and adaptive immune responses in asthma. *Nature medicine* 2012;18:673-83.

15. Holgate ST. Pathogenesis of asthma. *Clinical & Experimental Allergy* 2008;38:872-97.

16. Cookson W. The alliance of genes and environment in asthma and allergy. *Nature* 1999;402:B5-11.

17. Kleeberger SR, Peden D. Gene-environment interactions in asthma and other respiratory diseases. *Annu Rev Med* 2005;56:383-400.

18. Matthew M, Denise F, Shaun H, et al. The global burden of asthma: executive summary of the GINA Dissemination Committee report. *Allergy* 2004;59:469-78.

19. Bobolea I, Arismendi E, Valero A, et al. Early Life Origins of Asthma: A Review of Potential Effectors. *J Investig Allergol Clin Immunol* 2019;29:168-79.

20. Haland G, Carlsen KC, Sandvik L, et al. Reduced lung function at birth and the risk of asthma at 10 years of age. *N Engl J Med* 2006;355:1682-9.

21. Fuchs O, Bahmer T, Rabe KF, et al. Asthma transition from childhood into adulthood. *The Lancet Respiratory Medicine* 2017;5:224-34.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

22. Liu X, Olsen J, Agerbo E, et al. Maternal preeclampsia and childhood asthma in the offspring. *Pediatr Allergy Immunol* 2015;26:181-5.

23. Stokholm J, Sevelsted A, Anderson UD, et al. Preeclampsia Associates with Asthma, Allergy, and Eczema in Childhood. *Am J Respir Crit Care Med* 2017;195:614-21.

24. Mirzakhani H, Carey VJ, McElrath TF, et al. Impact of Preeclampsia on the Relationship between Maternal Asthma and Offspring Asthma. An Observati on from the VDAART Clinical Trial. *Am J Respir Crit Care Med* 2019;199:32-42.

25. Byberg KK, Ogland B, Eide GE, et al. Birth after preeclamptic pregnancies: association with allergic sensitization and allergic rhinoconju nctivitis in late childhood; a historically matched cohort study. *BMC Pediatr* 2014;14:101.

26. Wilmink FA, den Dekker HT, de Jongste JC, et al. Maternal blood pressure and hypertensive disorders during pregnancy and childhood respiratory morbidi ty: the Generation R Study. *Eur Respir J* 2018;52.

27. Shaheen SO, Macdonald-Wallis C, Lawlor DA, et al. Hypertensive disorders of pregnancy, respiratory outcomes and atopy in childhood. *Eur Respir J* 2016;47:156-65.

28. Li P, Xiong T, Hu Y. Hypertensive disorders of pregnancy and risk of asthma in offspring:

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

protocol for a systematic review and meta-analysis. *BMJ open* 2020;10:e035145.

29. Moher D, Liberati A, Tetzlaff J, et al. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *Ann Intern Med* 2009;151:264-9.

30. Stroup DF, Berlin JA, Morton SC, et al. Meta-analysis of observational studies in epidemiology: a proposal for reporting. Meta-analysis Of Ob servational Studies in Epidemiology (MOOSE) group. *Jama* 2000;283:2008-12.

31. Andreas S. Critical evaluation of the Newcastle-Ottawa scale for the assessment of the quality of nonrandomized studies in meta-analyses. *European Journal of Epidemiology* 2010;25:603-5.

32. Nafstad P, Samuelsen SO, Irgens LM, et al. Pregnancy complications and the risk of asthma among Norwegians born between 1967 and 1993. *Eur J Epidemiol* 2003;18:755-61.

33. Rusconi F, Galassi C, Forastiere F, et al. Maternal complications and procedures in pregnancy and at birth and wheezing phenotypes in children. *Am J Respir Crit Care Med* 2007;175:16-21.

34. Aspberg S, Dahlquist G, Kahan T, et al. Confirmed association between neonatal phototherapy or neonatal icterus and risk of childhood asthma Is neonatal phototherapy

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

associated with an increased risk for hospitalized childhood bronchial asthma. *Pediatr Allergy Immunol* 2010;21:e733-9.

35. Magnus MC, Haberg SE, Magnus P, et al. Pre-eclampsia and childhood asthma. *Eur Respir J* 2016;48:1622-30.

36. Duval S, Tweedie R. Trim and fill: a simple funnel-plot–based method of testing and adjusting for publication bias in meta-analysis. *Biometrics* 2000;56:455-63.

37. Moher D, Shamseer L, Clarke M, et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Systematic reviews* 2015;4:1. doi: 10.1186/2046-4053-4-1 [published Online First: 1 March 2015]

38. Wardenaar KJ, de Jonge P. Diagnostic heterogeneity in psychiatry: towards an empirical solution. *BMC medicine* 2013;11:1-3.

For peer review only

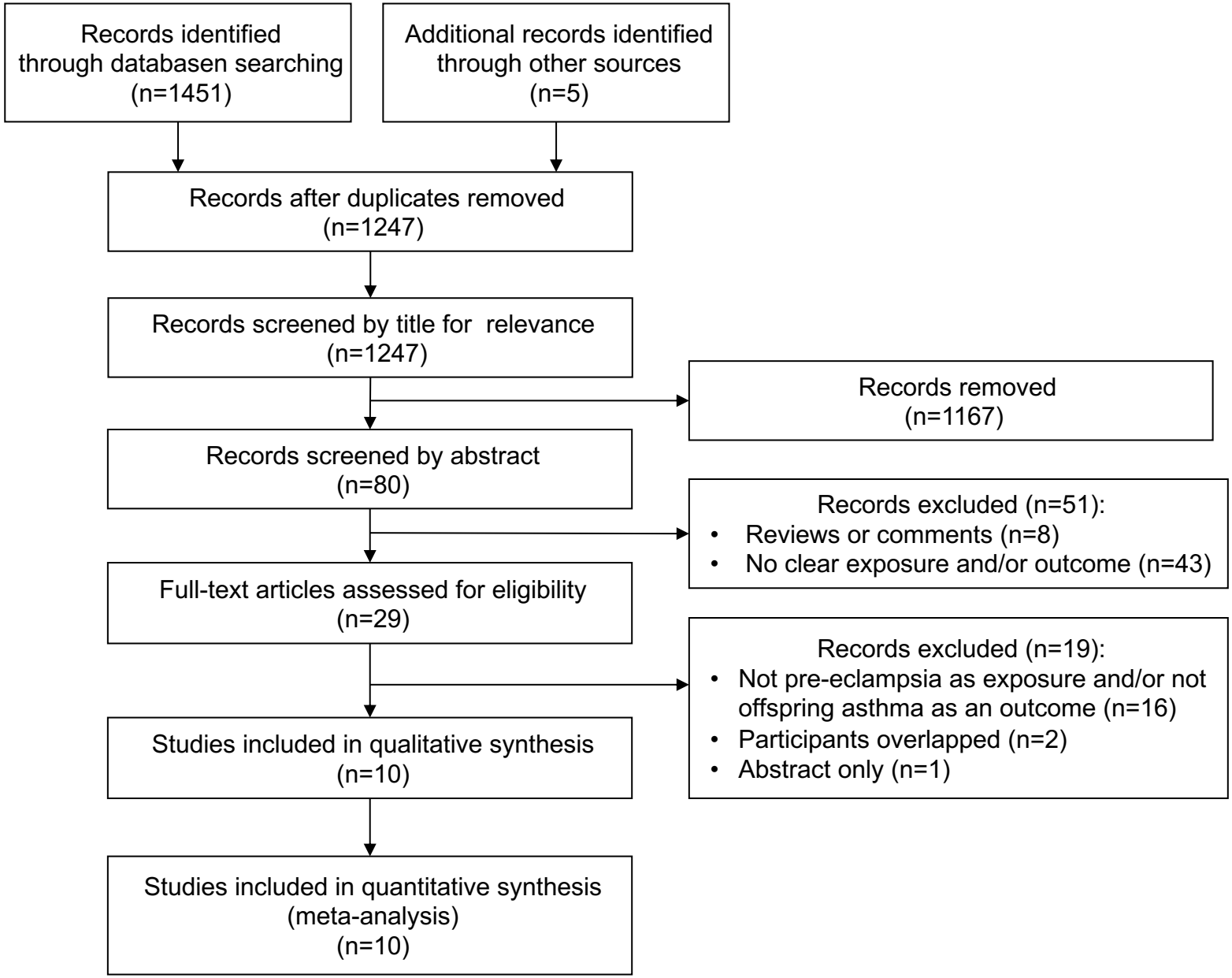
For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

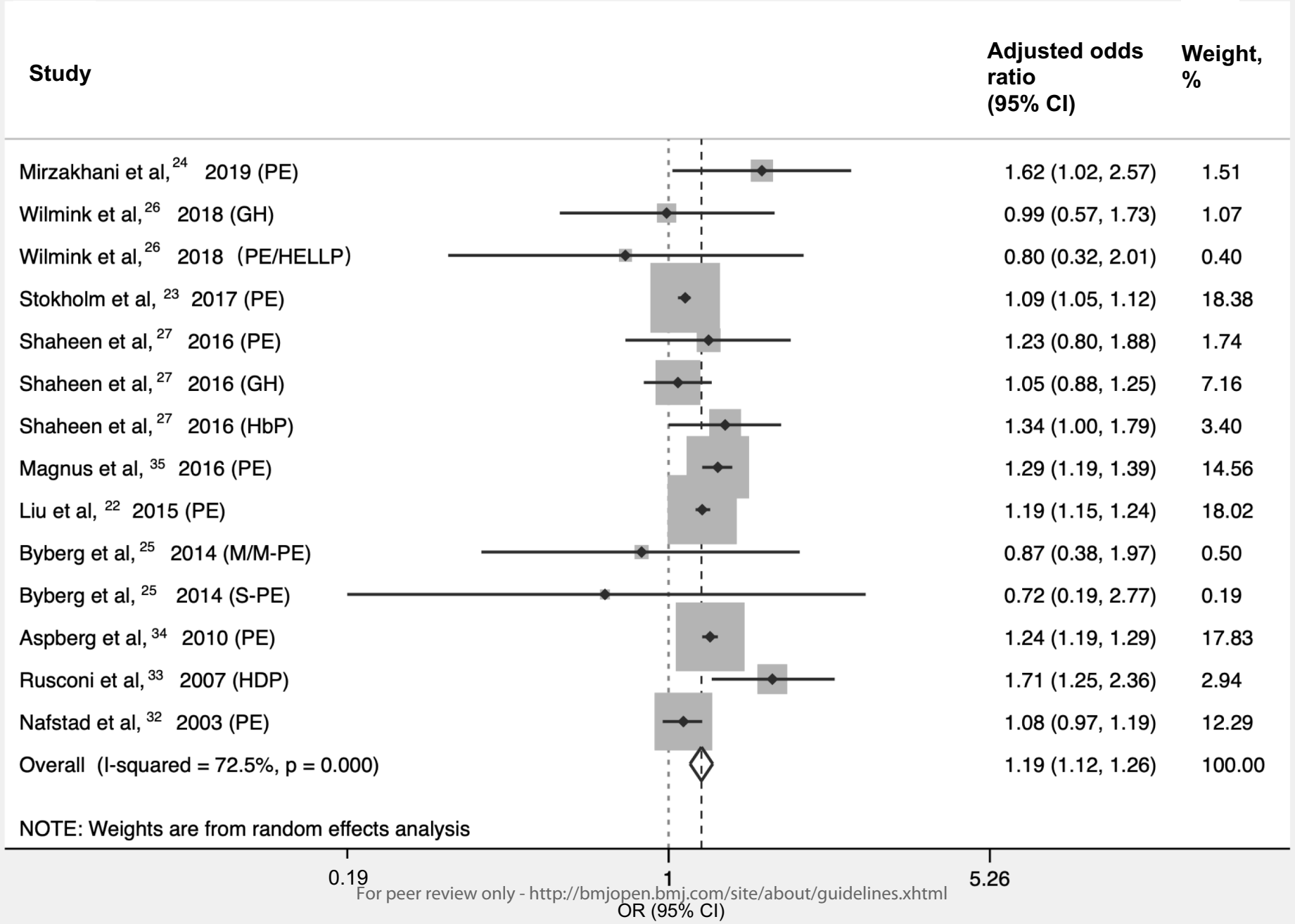
Figure 1. Flow diagram of the studies selected for inclusion in the systematic review.

Figure 2. Forest plot for studies on the association between hypertensive disorders in pregnancy (HDP) and asthma in offspring. Odds ratios (ORs) are calculated using random-effects analysis. Diamonds indicate effect size, size of markers, 95% CI. Abbreviations: PE, preeclampsia; GH, gestational hypertension; HELLP, hemolysis, elevated liver enzymes, and low platelets; HDP, hypertension disorder in pregnancy; HbP, hypertension before pregnancy; M/M-PE, mild/moderate preeclampsia; S-PE, severe preeclampsia; HDP, hypertension disorder in pregnancy; OR, odds ratio; CI, confidence interval.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml


Records identified through databasen searching (n=1451)
Additional records identified through other sources (n=5)
Records after duplicates removed (n=1247)
Records screened by title for relevance (n=1247)
Records removed (n=1167)
Records screened by abstract (n=80)
Records excluded (n=51):
• Reviews or comments (n=8)
• No clear exposure and/or outcome (n=43)
Full-text articles assessed for eligibility (n=29)
Records excluded (n=19):
• Not pre-eclampsia as exposure and/or not offspring asthma as an outcome (n=16)
• Participants overlapped (n=2)
• Abstract only (n=1)
Studies included in qualitative synthesis (n=10)
Studies included in quantitative synthesis (meta-analysis) (n=10)



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplementary Online-Only Tables and Figures

Supplemental eTable 1. Search strategy for each electronic database.

Supplemental eTable 2 Newcastle-Ottawa Scale (NOS) for cohort studies.

Supplemental eTable 3. Effect estimates and adjusted confounders in the included studies.

Supplemental eTable 4. Sensitivity analysis of the included studies.

Supplemental eFigure 1. Forest plot for studies on the association between subtypes of HDP and asthma in offspring.

Supplemental eFigure 2. Funnel plot of published studies on the association between HDP and asthma in offspring.

For peer review only

Supplemental eTable 1. Search strategy for each electronic database.

#1	Search terms	No of records returned
1	hypertensive disorders during pregnancy	
2	hypertensive disorders in pregnancy	
3	hypertensive disorders of pregnancy	
4	pregnancy-induced hypertension	
5	eclampsia	
6	pre-eclampsia	
7	preeclampsia	
8	preeclamptic pregnancy	
9	chronic hypertension	
10	gestational hypertension	
11	chronic hypertension with superimposed preeclampsia	
12	postpartum hypertension	
13	gravid hypertension	
14	maternal hypertensive disorders	
15	(#1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14)	
16	asthma	
17	wheeze	
18	wheezing	
19	shortness of breath	
20	bronchial spasm	
21	bronchospasm	
22	bronchoconstriction	
23	bronchoconstrict	
24	bronchial hyperreactivity	
25	bronchial hyperresponsiveness	
26	reactive airway disease	
27	obstructive lung disease	
28	air low limitation	
29	chronic obstructive respiratory disorder	
30	chronic obstructive pulmonary disease	
31	(#16 OR #17 OR #18 OR #19 OR #20 OR #21 OR #22 OR #23 OR #24 OR #25 OR #26 OR #27 OR #28 OR #29 OR #30)	
32	(#15 AND #31)	
33	limit #32 to English language	
34	limit #33 to humans	

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplemental eTable 2. Newcastle-Ottawa Scale (NOS) for cohort studies.

Study	Item & scores								Scores and Quality
	Representiveness of the exposed cohort（1）	Selection of the non-exposed cohort（1）	Ascertainment of exposure（1）	Demonstration that outcome of interest was not present at start of study（1）	Compare ability of cohorts on the basis of the design or analysis（2）	Assessment of outcome（1）	Was follow up long enough for outcomes to occur（1）	Adequacy of follow up of cohorts（1）	
Mirzakhani et al (2019)	0	1	1	1	2	0	0	1	6 (M)
Wilmink et al (2018)	1	1	1	1	1	0	1	1	7 (H)
Stokholm et al (2017)	1	1	1	1	2	1	1	1	9 (H)
Shaheen et al (2016)	1	1	1	1	2	0	1	1	8 (H)
Magnus et al (2016)	1	1	1	1	2	1	1	1	9 (H)
Liu et al (2015)	1	1	1	1	2	1	1	1	9 (H)
Byberg et al (2014)	0	1	1	1	2	0	1	1	7(H)
Aspberg et al (2010)	1	1	1	1	0	1	1	0	6 (M)
Rusconi et al (2007)	1	1	1	1	1	0	1	1	7(H)
Nafstad et al (2003)	1	1	1	1	0	1	1	1	7(H)

Note: The quality of the study was divided into 3 categories: high quality (scored 7–9), moderate quality (scored 4–6) and low quality (scored 0–3).

Abbreviations: M, moderate quality; H, high quality.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplemental eTable 3. Effect estimates and adjusted confounders in the included studies.

Author (year)	Effect estimate	Confounders adjusted					
		Mothers				Offspring	
		Maternal asthma	Maternal smoking during pregnancy	Maternal BMI	Others	Birthweight and gestational age	Others
Mirzakhani et al (2019)	adjusted RR [a]: asthma: 1.62 (1.02–2.57)	/	/	maternal BMI at first appointment	race, maternal age, trial arm, maternal baseline vitamin D, clinical center, maternal marital status, maternal education, gestational diabetes, mode of delivery, parity	gestational age, birthweight	sex
	adjusted HR [b]: asthma: 2.68 (1.30–5.61) asthma and recurrent wheeze: 4.73 (2.20–10.07)	yes	/	maternal BMI at first appointment	race, maternal age, trial arm, maternal baseline vitamin D, clinical center, maternal marital status, maternal education, gestational diabetes, mode of delivery, parity	gestational age, birthweight	sex
Wilmink et al (2018)	adjusted OR: current asthma [c]: 0.99 (0.57, 1.73) (GH); 0.80 (0.32, 2.01) (PE/ HELLP) current wheezing [d]: 0.95 (0.63, 1.42) (GH); 0.70 (0.36, 1.39) (PE/ HELLP)	/	yes	pre-pregnancy BMI	maternal age, maternal ethnicity, educational level, nulliparity, folic acid use	/	sex

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplemental eTable 3. Effect estimates and adjusted confounders in the included studies (continued).

Author (year)	Effect estimate	Confounders adjusted					
		Mothers				Offspring	
		Maternal asthma	Maternal smoking during pregnancy	Maternal BMI	Others	Birthweight and gestational age	Others
Stokholm et al (2017)	adjusted IRR: Registry-based cohort: 1.09 (1.05, 1.12)	/	/	/	mode of delivery, parity, multiple births, season of birth, maternal age, and mother's disease [e]	birth weight, gestational age	sex, age and calendar year, and season of birth
	adjusted OR [f]: COPSAC$_{2000}$ Birth Cohort: 1.13 (0.31, 4.13)	yes	/	/	type of delivery	/	/
Shaheen et al (2016)	adjusted OR: 1.20 (0.78, 1.83) (preeclampsia) 1.04 (0.87, 1.24) (GH) 1.33 (1.00, 1.78) (HbP)	yes	/	pre-pregnancy BMI	maternal age, maternal parity, maternal education, multiple pregnancy and maternal ethnicity	/	sex
	adjusted OR: 1.23 (0.80, 1.88) (preeclampsia) 1.05 (0.88, 1.25) (GH) 1.34 (1.00, 1.79) (HbP)	yes	yes	pre-pregnancy BMI	maternal age, maternal parity, maternal education, multiple pregnancy and maternal ethnicity	/	sex

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplemental eTable 3. Effect estimates and adjusted confounders in the included studies (continued).

Author (year)	Effect estimate	Confounders adjusted					
		Mothers				Offspring	
		Maternal asthma	Maternal smoking during pregnancy	Maternal BMI	Others	Birthweight and gestational age	Others
Magnus et al (2016)	adjusted RR: registry-based study: 1.31 (1.22, 1.41)	yes	/	/	maternal age, parity, education	/	sex
	adjusted RR: registry-based study: 1.29 (1.19–1.39)	yes	yes	/	maternal age, parity, education	/	sex
	adjusted OR: sibling pair analysis: 1.07 (0.87, 1.33)	yes	/	/	maternal age, parity, education	/	sex
	adjusted RR: MoBa study: 1.19 (0.99–1.44)	yes	yes	pre-pregnancy BMI	maternal age, parity, education	/	sex
Liu et al (2015)	adjusted IRR: nested case-control study: 1.19 (1.15, 1.24)	yes	yes	/	maternal age, maternal social status at birth, parity, and maternal country of origin	/	/
	adjusted IRR: case-sibling study: 1.06 (0.98, 1.14)	/	yes	/	maternal age, maternal social status at birth, and parity	/	sex, calendar year of birth

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplemental eTable 3. Effect estimates and adjusted confounders in the included studies (continued).

Author (year)	Effect estimate	Confounders adjusted					
		Mothers				Offspring	
		Maternal asthma	Maternal smoking during pregnancy	Maternal BMI	Others	Birthweight and gestational age	Others
Byberg et al (2014)	adjusted OR: FU1, asthma: 0.87 (0.38, 1.97) (mild/moderate preeclampsia) 0.72 (0.19, 2.77) (severe preeclampsia)	yes	yes	maternal BMI	caesarean section, maternal age	[g]	sex, being firstborn
	adjusted OR: FU2, current asthma [h]: 1.11 (0.45, 2.73) (mild/moderate preeclampsia) 0.69 (0.15, 3.18) (severe preeclampsia)	yes	yes	maternal BMI	caesarean section, maternal age	[g]	sex, being firstborn
Aspberg et al (2010)	adjusted OR: 1.24 (1.19–1.29)	/	yes	/	maternal age, parity, and involuntary childlessness	/	year of birth
Rusconi et al (2007)	adjusted OR: transient early wheezing [i]: 1.47 (1.14–1.89) persistent wheezing [j]: 1.71 (1.25–2.36) late-onset wheezing [k]: 1.45 (1.06–1.48)	/	yes	/	study center, parental atopy, maternal age, socioeconomic status, season of the questionnaire's completion, and person who completed the questionnaire	/	sex, low birth weight, siblings

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplemental eTable 3. Effect estimates and adjusted confounders in the included studies (continued).

Author (year)	Effect estimate	Confounders adjusted					
		Mothers				Offspring	
		Maternal asthma	Maternal smoking during pregnancy	Maternal BMI	Others	Birthweight and gestational age	Others
Nafstad et al[l] (2003)	adjusted OR: 1.08 (0.97–1.19) (total)	/	/	/	hyperemesis, maternal age, maternal education, plurality, single parent family, death or emigration during the study period, maternal atopy, diabetes mellitus, lower respiratory tract infections, and other maternal infections during pregnancy, other pregnancy complications [m]	/	sex, birth order, year of birth, geographical district of birth

Note:

[a] mothers with preeclampsia compared with mothers without preeclampsia;

[b] mothers with asthma with preeclampsia compared with mothers without asthma without preeclampsia;

[c] Current asthma was defined as physician diagnosed asthma ever, with either current wheezing or the use of inhaled medication in the past 12 months;

[d] Current wheezing, physician diagnosed asthma ever and use of inhaled medication (bronchodilators, corticosteroids) in the past 12 months was assessed by parental questionnaire at age 10 years;

[e] mother's disease, not mentioned;

[f] mothers with asthma with preeclampsia compared with mothers with asthma without preeclampsia (all the mothers had been diagnosed with asthma in the whole cohort);

[g] All the children were born full term;

[h] Current asthma at FU2 was defined as asthma ever, in addition to asthma symptoms or the use of asthma medication during the last 12 months;

[i] transient early wheezers: those who had at least one lower respiratory tract illness with wheezing (bronchiolitis or asthmatic bronchitis or asthma or wheezing) in the first 2 yr of life, and had no wheezing in the last 12 mo;

[j] persistent wheezers: those who had at least one lower respiratory tract illness with wheezing in the first 2 yr of life, and had wheezing in the last 12 mo;

[k] late-onset wheezers: those with no lower respiratory tract illness with wheezing in the first 2 yr of life but who had wheezing in the last 12 mo (late-onset wheezers).

[l] Only persons with severe asthma are registered in NIAR;

[m] pregnancy complications, including hemorrhage, anemia, cervix insufficiency, placenta dysfunction, rhesus immunization, preterm contractions.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Abbreviations: HDP, hypertension disorder in pregnancy; GH, gestational hypertension; HELLP, hemolysis, elevated liver enzymes, and low platelets; HbP, hypertension before pregnancy; VDRRT, Vitamin D Antenatal Asthma Reduction Trial; ISSHP, International Society for the Study of Hypertension in Pregnancy; ACOG, American College of Obstetricians and Gynaecologists; $COPSAC_{2000}$, Copenhagen Prospective Studies on Asthma in $Childhood_{2000;}$ ALSPAC, Avon Longitudinal Study of Parents and Children; DMBR, Danish Medical Birth Registry; ISAAC, International Study of Asthma and Allergies in Childhood; SIDRIA, Italian Studies of Respiratory Disorders in Childhood and the Environment; MBRN, Medical Birth Registry of Norway; NIAR, National Insurance Administration Register.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplemental eTable 4. Sensitivity analysis of the included studies.

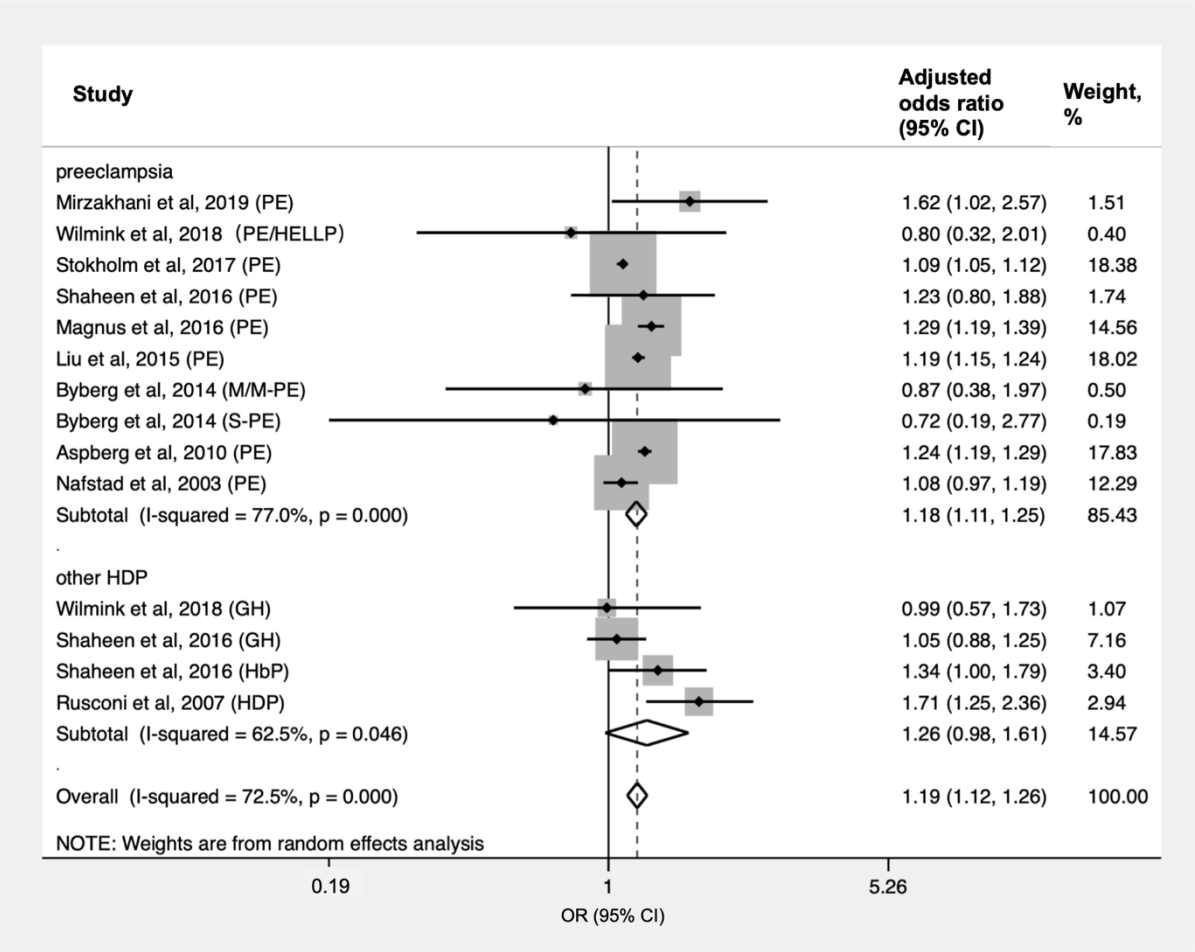
Study omitted	Estimate	95% CI
Mirzakhani et al, 2019 (PE)	1.18	1.11 - 1.25
Wilmink et al, 2018 (GH)	1.19	1.12 - 1.26
Wilmink et al, 2018 （PE/HELLP）	1.19	1.12 - 1.26
Stokholm et al, 2017 (PE)	1.21	1.15 - 1.27
Shaheen et al, 2016 (PE)	1.18	1.11 - 1.25
Shaheen et al, 2016 (GH)	1.20	1.12 - 1.26
Shaheen et al, 2016 (HbP)	1.18	1.11 - 1.23
Magnus et al, 2016 (PE)	1.16	1.09 - 1.23
Liu et al, 2015 (PE)	1.18	1.09 - 1.27
Byberg et al, 2014 (M/M-PE)	1.18	1.11 - 1.25
Byberg et al, 2014 (S-PE)	1.18	1.11 - 1.25
Aspberg et al, 2010 (PE)	1.17	1.10 - 1.24
Rusconi et al, 2007 (HDP)	1.17	1.11 - 1.24
Nafstad et al, 2003 (PE)	1.19	1.11 - 1.26

Abbreviations: HDP, hypertension disorder in pregnancy; GH, gestational hypertension; HELLP, hemolysis, elevated liver enzymes, and low platelets; HbP, hypertension before pregnancy; CI, confidence intervals.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplemental eFigure 1. Forest plot for studies on the association between subtypes of HDP and asthma in offspring.

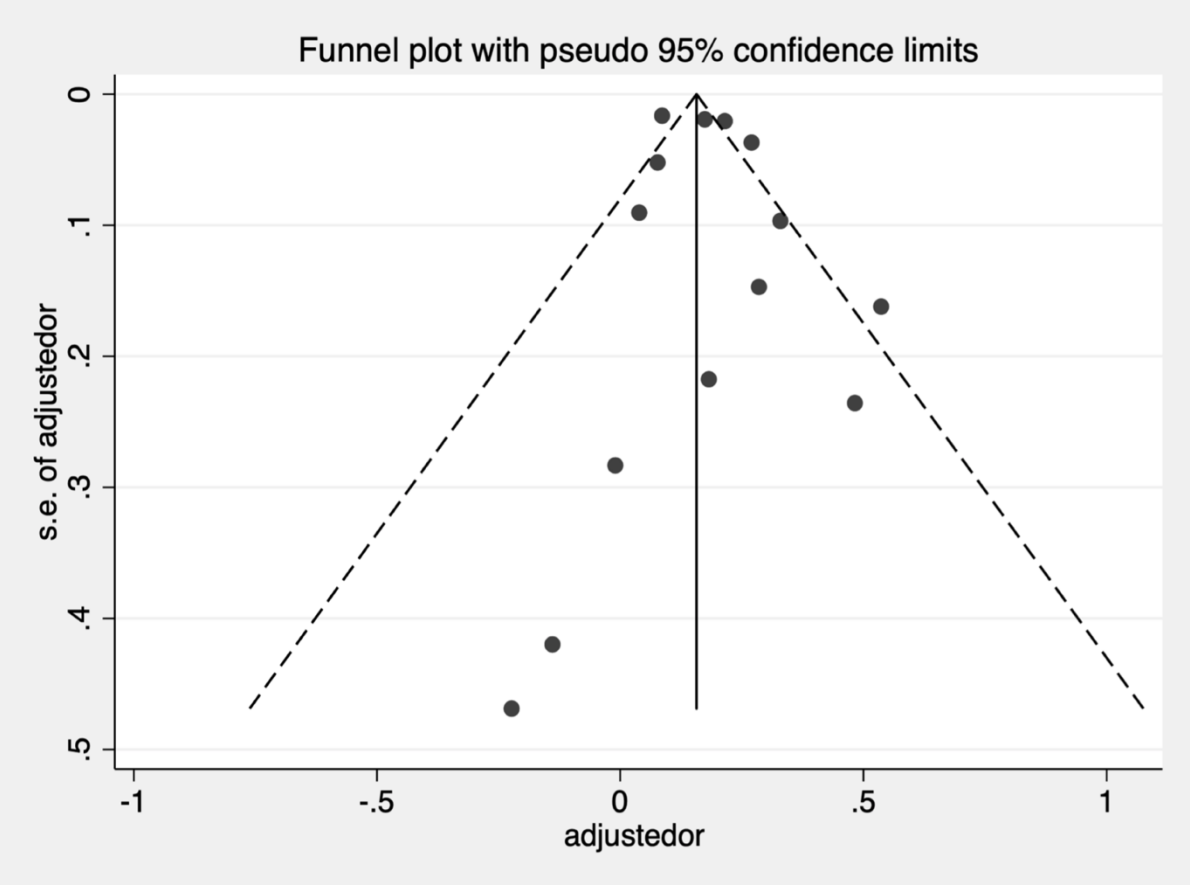


Forest plots displaying adjusted estimates examining the association between subtypes of HDP and asthma in offspring.

Abbreviations: PE, preeclampsia; GH, gestational hypertension; HELLP, hemolysis, elevated liver enzymes, and low platelets; HDP, hypertension disorder in pregnancy; HbP, hypertension before pregnancy; M/M-PE, mild/moderate preeclampsia; S-PE, severe preeclampsia; HDP, hypertension disorder in pregnancy; OR, odds ratio; CI, confidence intervals.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplemental eFigure 2. Funnel plot of published studies on the association between HDP and asthma in offspring.



Abbreviations: ORs, odds ratios

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

MOOSE Checklist for Meta-analyses of Observational Studies

Item No	Recommendation	Reported on Page No
Reporting of background should include		
1	Problem definition	5
2	Hypothesis statement	6
3	Description of study outcome(s)	6
4	Type of exposure or intervention used	5
5	Type of study designs used	7
6	Study population	6
Reporting of search strategy should include		
7	Qualifications of searchers (eg, librarians and investigators)	None
8	Search strategy, including time period included in the synthesis and key words	8
9	Effort to include all available studies, including contact with authors	8
10	Databases and registries searched	8
11	Search software used, name and version, including special features used (eg, explosion)	9
12	Use of hand searching (eg, reference lists of obtained articles)	8
13	List of citations located and those excluded, including justification	8,9
14	Method of addressing articles published in languages other than English	9
15	Method of handling abstracts and unpublished studies	9
16	Description of any contact with authors	8
Reporting of methods should include		
17	Description of relevance or appropriateness of studies assembled for assessing the hypothesis to be tested	8
18	Rationale for the selection and coding of data (eg, sound clinical principles or convenience)	8
19	Documentation of how data were classified and coded (eg, multiple raters, blinding and interrater reliability)	9
20	Assessment of confounding (eg, comparability of cases and controls in studies where appropriate)	10,11
21	Assessment of study quality, including blinding of quality assessors, stratification or regression on possible predictors of study results	9,10
22	Assessment of heterogeneity	11
23	Description of statistical methods (eg, complete description of fixed or random effects models, justification of whether the chosen models account for predictors of study results, dose-response models, or cumulative meta-analysis) in sufficient detail to be replicated	11
24	Provision of appropriate tables and graphics	10,11
Reporting of results should include		
25	Graphic summarizing individual study estimates and overall estimate	11
26	Table giving descriptive information for each study included	11
27	Results of sensitivity testing (eg, subgroup analysis)	16
28	Indication of statistical uncertainty of findings	None

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Item No	Recommendation	Reported on Page No
Reporting of discussion should include		
29	Quantitative assessment of bias (eg, publication bias)	18
30	Justification for exclusion (eg, exclusion of non-English language citations)	21
31	Assessment of quality of included studies	19
Reporting of conclusions should include		
32	Consideration of alternative explanations for observed results	23
33	Generalization of the conclusions (ie, appropriate for the data presented and within the domain of the literature review)	22
34	Guidelines for future research	23
35	Disclosure of funding source	23-24

From: Stroup DF, Berlin JA, Morton SC, et al, for the Meta-analysis Of Observational Studies in Epidemiology (MOOSE) Group. Meta-analysis of Observational Studies in Epidemiology. A Proposal for Reporting. *JAMA*. 2000;283(15):2008-2012. doi: 10.1001/jama.283.15.2008.

Transcribed from the original paper within the NEUROSURGERY® Editorial Office, Atlanta, GA, United Sates. August 2012.

For peer review only

BMJ Open

Hypertensive Disorders in Pregnancy and Risk of Asthma in Offspring: A Systematic Review and Meta-Analysis

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-046769.R1
Article Type:	Original research
Date Submitted by the Author:	01-Jul-2021
Complete List of Authors:	Li, Ping; West China Second University Hospital, Sichuan University, Department of Pediatrics; Key Laboratory of Birth Defects and Related Diseases of Women and Children (Sichuan University) Ministry of Education, China Xiong, tao; West China Second University Hospital, Sichuan University, Department of Pediatrics; Key Laboratory of Birth Defects and Related Diseases of Women and Children (Sichuan University) Ministry of Education, China, Hu, Yong; West China Second University Hospital, Sichuan University, Department of Pediatrics; Key Laboratory of Birth Defects and Related Diseases of Women and Children (Sichuan University) Ministry of Education, China
Primary Subject Heading:	Paediatrics
Secondary Subject Heading:	Cardiovascular medicine, Immunology (including allergy), Obstetrics and gynaecology
Keywords:	Hypertension < CARDIOLOGY, Asthma < THORACIC MEDICINE, PAEDIATRICS, Community child health < PAEDIATRICS, Fetal medicine < OBSTETRICS, PERINATOLOGY

SCHOLARONE™
Manuscripts



I, the Submitting Author has the right to grant and does grant on behalf of all authors of the Work (as defined in the below author licence), an exclusive licence and/or a non-exclusive licence for contributions from authors who are: i) UK Crown employees; ii) where BMJ has agreed a CC-BY licence shall apply, and/or iii) in accordance with the terms applicable for US Federal Government officers or employees acting as part of their official duties; on a worldwide, perpetual, irrevocable, royalty-free basis to BMJ Publishing Group Ltd ("BMJ") its licensees and where the relevant Journal is co-owned by BMJ to the co-owners of the Journal, to publish the Work in this journal and any other BMJ products and to exploit all rights, as set out in our licence.

The Submitting Author accepts and understands that any supply made under these terms is made by BMJ to the Submitting Author unless you are acting as an employee on behalf of your employer or a postgraduate student of an affiliated institution which is paying any applicable article publishing charge ("APC") for Open Access articles. Where the Submitting Author wishes to make the Work available on an Open Access basis (and intends to pay the relevant APC), the terms of reuse of such Open Access shall be governed by a Creative Commons licence – details of these licences and which Creative Commons licence will apply to this Work are set out in our licence referred to above.

Other than as permitted in any relevant BMJ Author's Self Archiving Policies, I confirm this Work has not been accepted for publication elsewhere, is not being considered for publication elsewhere and does not duplicate material already published. I confirm all authors consent to publication of this Work and authorise the granting of this licence.

For peer review only

Hypertensive Disorders in Pregnancy and Risk of Asthma in Offspring:

A Systematic Review and Meta-Analysis

Ping Li, MD,[1,2] Tao Xiong*, MD PhD [1,2,3], Yong Hu, MD [1,2]

[1] Department of Pediatrics, West China Second University Hospital, Sichuan University, Chengdu, China

[2] Key Laboratory of Birth Defects and Related Diseases of Women and Children (Sichuan University) Ministry of Education, China

[3] Deep Underground Space Medical Center, West China Hospital, Sichuan University, Chengdu, China

***Correspondence**

Dr. Tao Xiong

No. 20, Section Three, South Renmin Road, Chengdu, China

Tel.: 86 13730652940

Email: tao_xiong@126.com

Word count: 2925

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

ABSTRACT

Objectives The association between hypertensive disorders in pregnancy (HDP) and an increased risk of asthma in offspring remains controversial. No systematic review of this topic has been performed. The aim of this systematic review was to summarize the available evidence regarding the association between HDP and the risk of asthma in offspring.

Design Systematic review and meta-analysis.

Methods On the basis of a prepared protocol, a systematic search of PubMed, EMBASE, the Cochrane Library and Web of Science was performed using a detailed search strategy from the database inception to Jan 17, 2020. Cohort, case-control and cross-sectional studies published in English reporting the diagnoses of maternal HDP and asthma in offspring were included. The Meta-analysis of Observational Studies in Epidemiology (MOOSE) guidelines were followed throughout the study. The estimated pooled odds ratios (ORs) of HDP and asthma in offspring were calculated from the studies, and the meta-analysis was performed using random-effects models.

Results Ten cohort studies involving a total of 6,270,430 participants were included. According to the Newcastle-Ottawa Scale (NOS), the overall methodological quality was good

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

since 8 studies were of high quality and 2 studies were of moderate quality. After controlling for potential confounders, HDP was associated with a possible increased risk of asthma in offspring, with a pooled adjusted odds ratio (aOR) of 1.19 (95% CI, 1.12- 1.26). The subgroup analyses according to HDP subgroups, sibling design, study quality, study location, offspring ages, singleton status, exposure assessment, outcome assessment, and adjusted factors showed similar results.

Conclusions Exposure to HDP may be associated with an increased risk of asthma in offspring. Further research is needed to verify the results and determine whether the observed relationship is causal.

PROSPERO registration number CRD42020148250.

Keywords: hypertensive disorders in pregnancy, preeclampsia, blood pressure, asthma, offspring, childhood, meta-analysis

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Strengths and limitations of this study

- The studies included in this meta-analysis adopted a prospective cohort design, and ten studies were of high quality, which could eliminate various confounding factors.
- An extremely large sample size of more than six million participants was included; thus, our findings are potentially more robust than those of any individual study.
- This study was based on a pre-prepared protocol and followed meta-analysis guidelines and the Meta-analysis of Observational Studies in Epidemiology guidelines.
- The diagnostic criteria for hypertensive disorders in pregnancy and asthma varied among the included studies, which may increase clinical heterogeneity.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

INTRODUCTION

Hypertensive disorders in pregnancy (HDP), which are estimated to affect 5% to 15% of all pregnancies, are recognized as among the most common obstetrical complications.[1 2] HDP are defined as any hypertension condition (systolic blood pressure (BP) ≥140 and/or diastolic BP ≥90 mmHg) that manifests before gestation or before 20 weeks and hypertension starting at or after 20 weeks.[3] There are mainly four subtypes of HDP, including preeclampsia-eclampsia, chronic hypertension, chronic hypertension with superimposed preeclampsia, and gestational hypertension (GH).[4] All HDP subtypes pose serious threats to the health of the offspring. Recent studies have confirmed that HDP are closely associated with a risk of offspring diseases, including lung disease,[5] high blood pressure,[6] congenital heart defects,[7] obesity,[8] autism spectrum disorder, attention-deficit/hyperactivity disorder, low cognitive function, anxiety/depression, and other neurodevelopmental disorders.[9 10]

Theoretically, HDP induce a detrimental in utero environment, causing systemic inflammation and oxidative stress in the vulnerable fetal lung. Therefore, the development of the fetal lung may be disturbed by abnormal inflammation and immune function. These disturbances may persist through late life in offspring, leading to subsequent respiratory

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

diseases. HDP may have great impacts on the fetal airway structure and respiratory and immune system development during the prenatal period, significantly increasing the susceptibility of offspring to asthma.[5 11] Asthma, which affects 1-18% of the population, is the most common lung disease associated with abnormal inflammation and immune function.[12 13] Asthma is characterized by variable symptoms, such as wheezing, shortness of breath, chest tightness and/or cough, and variable expiratory airflow limitation.[14-16] Since asthma can cause recurrent school absences, emergency department visits and hospitalizations, it has been regarded as a major public health challenge worldwide.

The etiology of asthma is not completely clear. In addition to genetic and environmental factors,[17 18] it is widely accepted that asthma originates early in life.[19] Negative events during the perinatal period increase the risk of asthma and impair lung function later in life.[20 21] Asthma symptoms beginning in adulthood may have originated in childhood.[22] Recurrent wheezing and other asthma-like symptoms usually begin as early as the first few weeks or months after birth, highlighting the need for the early recognition of and interventions for the risk factors of asthma to minimize the development of asthma and its potential long-term sequelae. Recent epidemiological studies have reported that maternal HDP may be a

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

potential risk factor for asthma in offspring.[23-25] However, evidence of the association between maternal HDP and asthma in offspring is inconsistent.[26-28] Hence, clarifying the magnitude of this association is extremely important. Therefore, we performed the current systematic review and meta-analysis to synthesize the available evidence regarding the relationship between HDP and asthma in offspring.

METHODS

Search strategy

We followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines, [29] Meta-analysis of Observational Studies in Epidemiology (MOOSE) guidelines [30], and a prepared protocol registered in the international prospective register of systematic reviews (PROSPERO, ID: CRD42020148250) and published in BMJ OPEN. [31]

Two reviewers (P.L. and T.X.) independently conducted a systematic search of the literature in PubMed, EMBASE, the Cochrane Library and Web of Science from database inception to Jan. 17, 2020. Search terms related to HDP and asthma were combined according to the principles of Boolean logic (using AND, OR, or NOT). The full search strategy is included in

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

eTable 1 in the Supplement. In addition, the reference lists of the included studies were manually searched to further identify eligible studies. The authors further requested the full text or research data of articles with only abstracts or unpublished articles. The authors of 3 studies were contacted for additional information. The searches were limited to human studies published in English. No restrictions were placed on the publication date, location of the study, or the age of the participants.

Study selection

Two investigators (P.L. and Y.H.) independently inspected the titles and abstracts of the citations and obtained the full texts. When consensus regarding eligibility could not be reached, a third reviewer (T.X.) was involved in the discussion. The eligibility criteria for inclusion in the systematic review included the following: 1) cohort, case-control, or cross-sectional studies in which a diagnosis of HDP was reported, asthma was the primary outcome of interest, and the secondary outcomes consisted of recurrent wheezing/wheezing, lung/pulmonary function, the severity of asthma (active asthma or asthma exacerbations) or asthma treatment; 2) studies in which the association between HDP and asthma was a main

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

objective of the study; 3) studies that contained original data and confirmed HDP through medical records and/or physician-diagnosed self-reporting; and 4) peer-reviewed published studies. The exclusion criteria were as follows: 1) case series, case reports, conference abstracts, letters to the editor, reviews and commentary articles; 2) studies with overlapping data (same data from the same population in two or more studies); and 3) studies without raw data (no data were provided for the meta-analysis).

Data extraction

Two reviewers (P.L. and Y.H.) independently extracted the data from the eligible studies using a standardized data extraction form. The titles and abstracts were retrieved from each article, stored and managed using EndNote reference manager. From each included study, we extracted the following information: first author's last name, year of publication, study location, study design, sample size, exposure (HDP and its subgroups), outcome diagnostic criteria, offspring age at diagnosis, adjusted/matched confounding variables and effect estimates [risk ratios (RRs) or odds ratios (ORs)] with the 95% confidence intervals (CIs). Any discrepancies were resolved by consensus with a third reviewer (T.X.).

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Bias and quality assessment

The quality assessments of the included studies were conducted by 2 reviewers (P.L. and Y.H.) independently, and any discrepancies were resolved by a third reviewer (T.X). The quality assessment followed our published protocol. [31] The quality of the cohort studies was assessed using the Newcastle-Ottawa Scale (NOS).[32] The study quality was classified into the following 3 categories: high quality (scores 7-9), moderate quality (scores 4-6) and low quality (scores 0-3).

Statistical analysis

The data were analyzed using Stata version 14.0 (StataCorp, College Station, TX, USA). The ORs and 95% CIs were used as a measure of the association between HDP and the risk of asthma in offspring among the studies. We computed a summary OR and its 95% CIs using the study-specific most-adjusted OR or RR (maternal HDP exposure compared with maternal non-HDP exposure) and its 95% CIs. We used a random-effects model to calculate the pooled ORs. The I^2 statistic (significance level, >50%) was applied to examine heterogeneity.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Forest plots were constructed to show the study-specific OR estimates and the pooled OR estimate. A protocol-based subgroup analysis was performed. [31] Since some potential confounding factors may exist, we performed detailed subgroup analyses to reduce the effects of confounding factors. Because the family situation and genetic factors are similar in siblings, we performed a subgroup analysis of sibling-design studies. Finally, subgroups analyses stratified by HDP subgroups, sibling design, study quality, study location, offspring ages, singleton status, exposure assessment, outcome assessment, and adjusted factors were performed. The sensitivity analysis was performed by omitting one study at a time to evaluate the changes in the pooled OR. The potential risk of publication bias was estimated by inspection of a funnel plot. Publication bias was further assessed by Egger's test and Begg's test (significance level, $P<0.05$) and the "trim-and-fill" procedure.

Patient and public involvement

Patients or the public were not involved.

RESULTS

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Literature search and selection

The original search produced 1,247 unique results after the removal of duplicates. Of these articles, twenty-nine full-text articles were reviewed after screening the titles and abstracts. Of these articles, nineteen articles were excluded because they did not meet the inclusion criteria. Finally, ten studies[23-28 33-36] were included in this study. The reasons for exclusion are outlined in Figure 1.

Study characteristics

The characteristics of the 10 included studies are shown in Table 1. The studies were published between 2003 and 2019. All selected studies[23-28 33-36] were cohort studies, and

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Table 1. Characteristics of the selected studies included in the systematic review and meta-analysis.

Author (year)	Study design	Country	Data source	Sample size	Ascertainment of HDP	Ascertainment of asthma	Offspring age at diagnosis	Study quality
Mirzakhani et al (2019) [25]	cohort	United States	VDAART	67 (mothers with preeclampsia) 739 (mothers without preeclampsia) 28 (mothers with asthma and preeclampsia) 294 (mothers with asthma without preeclampsia)	medical records (ACOG)	maternal-reported (physician-diagnosed)	3 yrs	M
Wilmink et al (2018) [27]	cohort	Netherlands	Generation R Study	206 (mothers with GH) 91 (mothers with preeclampsia or HELLP syndrome) 4,783 (control)	medical records (ISSHP，ACOG)	maternal-reported (physician-diagnosed)	10 yrs (8.6~12 yrs)	H
Stokholm et al (2017) [24]	cohort	Denmark	Registry-based cohort	62,728 (mothers with preeclampsia) 1,635,910 (mothers without preeclampsia)	medical records (ICD-8, ICD-10)	medical records (ICD-10)	0~15 yrs	H
			$COPSAC_{2000}$ Birth Cohort	20 (mothers with asthma and preeclampsia) 317 (mothers with asthma without preeclampsia)	medical records (ICD-10)	medical records (pediatrician-diagnosed)	7 yrs	
Shaheen et al (2016) [28]	cohort	UK	ALSPAC	161 (mothers with preeclampsia) 1,140 (mothers with GH) 308 (mothers with HbP) 6,486 (normotensive)	medical records (ISSHP criteria14)	maternal-reported (doctor-diagnosed)	7 yrs	H
Magnus et al (2016) [36]	cohort	Norway	Registry-based study	16,329 (mothers with preeclampsia) 390,578 (control)	medical records	medical records	7 yrs	H
Liu et al (2015) [23]	cohort	Denmark	DMBR	nested case-control study: 30,767 (mothers with preeclampsia) 1,239,965 (mothers without preeclampsia) case-sibling study: 3,608 (mothers with preeclampsia) 143,704 (mothers without preeclampsia)	medical records (ICD-8, ICD-10)	medical records (ICD-10)	3~7 yrs	H
Byberg et al (2014) [26]	cohort	Norway	Stavanger study	FU1 (F 10.8 yrs, M 11.8 yrs): 214 (mothers with preeclampsia) 366 (mothers without preeclampsia) FU2 (F, M: 12.8 yrs): 169 (mothers with preeclampsia) 279 (mothers without preeclampsia)	medical records	FU1:maternal-reported (physician-diagnosed) FU2: self-reported (ISAAC)	10~12 yrs	H
Aspberg et al (2010) [35]	cohort	Switzerland	Registry-based study	41,389 (mothers with preeclampsia) 1,296,083 (control)	medical records	medical records	2~15 yrs	M
Rusconi et al (2007) [34]	cohort	Italy	SIDRIA	862 (mothers with hypertension or preeclampsia) 14,729 (control)	medical records	maternal-reported	6~7 yrs	H
Nafstad et al (2003) [33]	cohort	Norway	Registry-based study	98,188 (mothers with preeclampsia) 1,428,341 (control)	medical records (ICD-8)	derived from NIAR (doctor-diagnosed, ICD-7, ICD-9)	7.5~25.5 yrs	H

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Abbreviations: HDP, hypertension disorders in pregnancy; GH, gestational hypertension; HELLP, hemolysis, elevated liver enzymes, and low platelets; HbP, hypertension before pregnancy; VDRRT, Vitamin D Antenatal Asthma Reduction Trial; ISSHP, International Society for the Study of Hypertension in Pregnancy; ACOG, American College of Obstetricians and Gynaecologists; $COPSAC_{2000}$, Copenhagen Prospective Studies on Asthma in $Childhood_{2000;}$ ALSPAC, Avon Longitudinal Study of Parents and Children; DMBR, Danish Medical Birth Registry; ISAAC, International Study of Asthma and Allergies in Childhood; SIDRIA, Italian Studies of Respiratory Disorders in Childhood and the Environment; MBRN, Medical Birth Registry of Norway; NIAR, National Insurance Administration Register; H, high quality (scores 7–9); M, median quality (scores 4–6).

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

there was a total of 6,270,430 participants. The sample sizes of the studies ranged from 580 to 1,698,638, and 4 studies[23 24 33 35] had sample sizes of more than 1 million. Nine studies[23 24 26-28 33-36] were performed in Europe, and one study[25] was performed in the United States. Four studies[28 33 34 36] included children aged 6 to 7 years, and two studies[26 27] included children aged 10 to 12 years. In 2 cohort studies,[23 36] the authors used a sibling analysis in which children with the same mother were included to control for shared genetic and social confounders. According to the NOS, the overall methodological quality was good since 8 studies[23 24 26-28 33 34 36] were of high quality and 2 studies[25 35] were of moderate quality (Supplemental eTable 2). The confounders that were adjusted in the studies are shown in eTable 3 in the Supplement.

Results of the meta-analyses

Primary outcome: HDP and risk of asthma in offspring

Ten studies[23-28 33-36] in which a diagnosis of HDP was reported and asthma was the outcome of interest were identified. Seven studies[24-28 33 36] described that the incidence of HDP ranged from 2.0% to 14.0% (mean, 6.7%; median, 5.6%; interquartile range [IQR], 3.3%-10.0%), and

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

an average of 4.6% mothers had preeclampsia (IQR, 2.3%-6.9%). The incidence of asthma ranged from 4.9% to 25.5% (mean, 14.7%; median, 12.5%; IQR, 6.5%-23.4%) in the offspring of mothers diagnosed with HDP and 3.6%-17.5% (mean, 8.9%; median, 8.6%; IQR, 4.8%-13.3%) in the controls.[23-28 34-36] Fourteen estimates from 10 unique studies included adjusted estimates and were included in the meta-analysis. Figure 2 displays the adjusted estimates, producing a pooled adjusted odds ratio (aOR) of 1.19 (95% CI, 1.12- 1.26), suggesting an increased risk of asthma among the offspring of mothers with HDP. Statistically significant heterogeneity was found across the studies (I^2 = 72.5%).

Secondary outcomes

Only a few included studies reported secondary outcomes, including recurrent wheezing/wheezing, active asthma, asthma exacerbations or asthma treatment. Only two studies reported the physiological parameters of lung function. One study reported a decreased forced expiratory volume in one second (FEV_1), and another study reported a reduced forced expiratory flow between 25% and 75% (FEF_{25-75}) in offspring of mothers with preeclampsia.[26 28]

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Subgroup and sensitivity analyses

We performed subgroup analyses to explore heterogeneity according to subgroups of HDP, study quality, study location, offspring ages, singleton births, exposure assessment, outcome assessment, and adjusted factors. The results of the subgroup analyses are outlined in Table 2. The subgroup analysis examining the associations between subtypes of HDP and asthma in offspring resulted in aORs of 1.18 (95% CI, 1.11-1.25) for preeclampsia and 1.26 (95% CI, 0.98-1.61) for other HDP (Supplemental eFigure 1). The sibling analysis failed to reveal a relationship between preeclampsia and asthma in offspring, with a pooled aOR of 1.06 (95% CI, 0.99-1.14). The high quality studies yielded a significant aOR of 1.17 (95% CI, 1.09-1.25). In addition, significant aORs were observed in studies conducted in both Europe and the United States. Regarding the offspring ages, the aORs in three age groups were significant, but not among the offspring aged 10-12 years. The aOR in the singleton subgroup was slightly higher than that in the nonsingleton subgroup, with decreased heterogeneity of I^2 = 47.2%. The aORs of any type of exposure assessment (HDP from self-reports or medical records) and any type of outcome assessment (asthma from maternal reports or medical

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

239 records) were all significant. In the subgroup analyses by adjusted factors, the aORs were

240 generally significant after adjusting for maternal

Table 2. Pooled ORs and heterogeneity of the subgroup analyses.

Variable	No. of Studies (No. of estimates)	OR (95%CI)	I^2 (%)	*P* for Heterogeneity
Category of HDP				
Preeclampsia	9 (10)	1.18 (1.11-1.25)	77.0	<0.001
Other	3 (4)	1.26 (0.98-1.61)	62.5	0.046
Sibling design (only preeclampsia)				
Yes	2 (2)	1.06 (0.99-1.14)	0.0	0.935
No	7 (8)	1.15 (1.05-1.26)	75.8	<0.001
Study quality				
Moderate	2 (2)	1.28 (1.08-1.51)	21.6	0.259
High	8 (12)	1.17 (1.09-1.25)	66.9	<0.001
Location (only preeclampsia)				
Europe	8 (9)	1.17 (1.10-1.25)	78.5	<0.001
United States	1 (1)	1.62 (1.02-2.57)	0.0	NA
Offspring ages (years)				
6-7	4 (6)	1.23 (1.10-1.38)	55.6	0.046
10-12	2 (4)	0.90 (0.61-1.33)	0.0	0.962
3	1 (1)	1.62 (1.02-2.57)	NA	NA
Others [a]	3 (3)	1.17 (1.08-1.27)	92.5	<0.001
Only singletons				
Yes	6 (7)	1.24 (1.18-1.31)	47.2	0.078
No	4 (7)	1.10 (1.06-1.13)	0.0	0.799
Exposure assessment				
Self-report	1 (1)	1.71 (1.25-2.35)	NA	NA
Medical records	9 (13)	1.18 (1.11-1.25)	70.1	<0.001
Outcome assessment				
Maternal-report	4 (7)	1.26 (1.05-1.52)	42.0	0.111
Medical records	6 (7)	1.18 (1.11-1.26)	82.8	<0.001
Maternal smoking during pregnancy				
Adjusted	7 (11)	1.23 (1.17-1.28)	30.7	0.154
Not adjusted	3 (3)	1.10 (1.03-1.17)	29.9	0.240
Maternal asthma				
Adjusted	5 (8)	1.22 (1.12-1.33)	50.4	0.049
Not adjusted	5 (6)	1.17 (1.05-1.29)	84.5	<0.001

241 [a] Others include 0 ~ 25.5 years.

242 Abbreviations: OR, odds ratio; CI, confidence intervals; HDP, hypertension disorders in pregnancy;

243 NA, not applicable.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

smoking during pregnancy (aOR, 1.23; 95% CI, 1.17-1.28) and maternal asthma (aOR, 1.22; 95% CI, 1.12-1.33). Meanwhile, the heterogeneity obviously decreased after adjusting for maternal smoking during pregnancy (I^2 = 30.7%) and maternal asthma (I^2 = 50.4%). The sensitivity analysis, which was performed by omitting one study at a time, showed that the overall pooled adjusted ORs did not substantially vary (Supplemental eTable 4).

Publication bias

A visual inspection of the funnel plot revealed potential publication bias (Supplemental eFigure 2), although both Egger's test and Begg's test were not statistically significant (Egger's test, *P* = 0.709; 95% CI, -1.26-1.80; Begg's test, *z* = 0.770 (continuity corrected), *Pr* > |*z*| = 0.443 (continuity corrected)). Therefore, we further applied the trim-and-fill method to adjust for the funnel plot asymmetry,[37] but the results showed that no trimming was performed, and the data were unchanged (''no trimming performed, data unchanged''). There was no indication of publication bias using Duval's trim-and-fill method (no new studies added).

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

DISCUSSION

In this systematic review and meta-analysis based on 10 observational studies involving 6,270,430 participants, we found that maternal HDP exposure was associated with approximately 20% increased odds of asthma in offspring compared with nonexposure. Most subgroup analyses showed that offspring have a slightly increased asthma risk after maternal HDP exposure. The results of the subgroup analysis examining the association between preeclampsia and asthma in offspring provided a significant aOR of 1.18.[23-28 33 35 36]

High heterogeneity was detected in this meta-analysis. Therefore, we performed subgroup analyses to explore the cause of the heterogeneity. Notably, we found that the heterogeneity significantly decreased when studies that considered only singletons and maternal smoking during pregnancy as a confounding factor were included, revealing that the number of offspring and maternal smoking during pregnancy may have contributed to the heterogeneity. Another explanation for the observed heterogeneity may be due to variability in the incidence of asthma in the offspring among the included studies. The incidence of asthma ranged from 4.9% to 25.5% in the offspring of mothers diagnosed with HDP and 3.6%-17.5% in the controls.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

The genetic susceptibility of offspring to asthma was evaluated through three analyses in this study. First, since the family situation and genetic factors are similar in siblings, we performed a subgroup analysis of two- sibling-design studies[23 36] and found that there was a borderline significant relationship between preeclampsia and asthma in the offspring (aOR, 1.06; 95% CI, 0.99-1.14). Second, a positive relationship (aOR, 1.22; 95% CI, 1.12-1.33) was observed between HDP and asthma in offspring after adjusting for maternal asthma.[23 25 26 28 36] The discrepancy between the two analyses may be due to the limited sample size or other unadjusted factors. The fundamental role of genetics should be noted in future studies evaluating maternal HDP and asthma in offspring.

Strengths and limitations

This systematic review has several strengths. First, this study had an extremely large sample size involving more than six million participants. Second, all ten studies adopted prospective cohort designs, and eight of the ten studies were of high quality. Five studies included birth registry-based cohorts.[23 24 33 35 36] Third, the estimates in the original studies were carefully adjusted for potential confounders. Fourth, our study was based on a preprepared protocol and followed the PRISMA guidelines and MOOSE guidelines.[29-31] Thus,

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

our meta-analysis provided reliable evidence showing a significantly increased risk of asthma in offspring exposed to maternal HDP. However, there are several potential limitations in this review. First, the included studies were limited to the English language, and potential non-English-language studies may have been missed. Second, the generalizability of the results worldwide should be performed cautiously because of the geographic restriction of the included studies, which mostly involved participants from Europe or the United States. Future clinical investigations performed in other regions are needed. Third, the different assessments of HDP and asthma used among the studies may potentially introduce misclassification bias. The diagnostic criteria for HDP and asthma varied among the included studies, which may increase clinical heterogeneity.[38] Several studies[23 33 35 36] followed the diagnosis of asthma according to the National Insurance Administration Register or National Prescribed Drug Register, which might exclude offspring with mild asthma. Fourth, important confounding factors were not fully adjusted in the included studies. Therefore, although a significant association exists between HDP and asthma in offspring, future research is needed to identify a comprehensive set of confounders to assess whether this association is causal or attributable to residual or unmeasured confounding. Although there was a significant influence

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

of HDP on asthma in offspring, the effects of HDP on the physiological parameters of lung function, asthma exacerbations, or asthma treatment need further consideration in future studies.

Exposure to maternal HDP in early life may be associated with an increased risk of developing asthma later in life. HDP appears to be an interesting predictor of asthma in offspring. This finding provides important evidence of the long-term consequences of exposure to maternal HDP on offspring. Furthermore, this study supports the fetal origin hypothesis of the development of asthma. Because maternal HDP is a common risk factor for multiple poor outcomes in offspring, early surveillance and interventions are warranted for high-risk children of mothers with HDP. HDP increase the risk of offspring diseases, which start from the fetus (such as preterm delivery and stillbirth) and may last to adulthood (multiorgan diseases, such as cardiovascular, endocrine, and neurodevelopmental functions). Early surveillance from the fetus stage to adulthood should be comprehensive. In addition, assessing whether maternal HDP was present when evaluating infants for asthma could be important. For future asthma studies, maternal HDP should be considered toa potential risk factor.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

CONCLUSION

Our study found a mild increased risk of asthma in offspring of mothers with HDP; this reliable result was derived from a large population and mostly high-quality studies. Additional robust research is needed to address the key limitations in the literature, use the gold standard for disease diagnosis, verify the results and determine whether the observed relationship is causal. We advocate for improved pediatric surveillance of infants exposed to maternal HDP as increased surveillance may be beneficial for the early diagnosis of and interventions for asthma.

Acknowledgments

We acknowledge the suggestions by Imti Choonara (Emeritus Professor in Child Health and Editor in Chief of BMJ Paeds Open, University of Nottingham, the Medical School, Derby, UK) regarding the conception and design of the analysis and other valuable advice regarding this manuscript.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Contributors

This study was conceived by P.L. and T.X. P.L., T.X., and Y.H. developed the eligibility criteria, search strategy, methodological quality assessment, data extraction and data summary plan. P.L. and T.X. wrote the protocol. T.X. supervised the work. All authors approved the final version.

Funding

This work is supported by the National Science Foundation of China (No. 81300525 to Tao Xiong and No. 81701540 to Ping Li); the Science and Technology Bureau of Sichuan Province (2020YJ0298 and 2020YFS0109); Health Commission of Sichuan Province (20PJ071); Deep Underground Space Medical Center (DUGM201809); and the Fundamental Research Funds for the Central Universities (SCU2019C4005).

Competing interests

None declared.

Patient consent for publication

Not required.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Provenance and peer review

Not commissioned; externally peer reviewed.

Data sharing statement

No data sharing available.

Ethics approval statement

Not required. This study does not involve human participants.

REFERENCES

1. Gillon TE, Pels A, Dadelszen PV, et al. Hypertensive disorders of pregnancy: A systematic review of international clinical practice guidelines. *Plos One* 2014;9:122.

2. Olson-Chen C, Seligman NS. Hypertensive Emergencies in Pregnancy. *Critical Care Clinics* 2016;32:29-41.

3. Brown MA, Magee LA, Kenny LC, et al. Hypertensive Disorders of Pregnancy: ISSHP Classification, Diagnosis, and Management Recommendations for International Practice.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Hypertension 2018;72:24-43.

4. Obstetriciansgynecologists ACO. Hypertension in pregnancy. Report of the American College of Obstetricians and Gynecologists' Task Force on Hypertension in Pregnancy. *Obstetrics & Gynecology* 2013;122:1122-31.

5. Harding R, Maritz G. Maternal and fetal origins of lung disease in adulthood. *Semin Fetal Neonatal Med* 2012;17:67-72.

6. Pinheiro T, Brunetto S, Ramos J, et al. Hypertensive disorders during pregnancy and health outcomes in the offspring: a systematic review. *Journal of developmental origins of health and disease* 2016;7:391-407.

7. Boyd HA, Basit S, Behrens I, et al. Association Between Fetal Congenital Heart Defects and Maternal Risk of Hypertensive Disorders of Pre gnancy in the Same Pregnancy and Across Pregnancies. *Circulation* 2017;136:39-48.

8. Zheng JS, Liu H, Ong KK, et al. Maternal Blood Pressure Rise During Pregnancy and Offspring Obesity Risk at 4 to 7 Years Old: The Jia xing Birth Cohort. *J Clin Endocrinol Metab* 2017;102:4315-22.

9. Tuovinen S, Eriksson JG, Kajantie E, et al. Maternal hypertensive pregnancy disorders

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

and cognitive functioning of the offspring: a systematic re view. *J Am Soc Hypertens* 2014;8:832-47.

10. Maher GM, O'Keeffe GW, Kearney PM, et al. Association of Hypertensive Disorders of Pregnancy With Risk of Neurodevelopmental Disorders in Offsp ring: A Systematic Review and Meta-analysis. *JAMA Psychiatry* 2018;75:809-19.

11. Postma DS, Bush A, van den Berge M. Risk factors and early origins of chronic obstructive pulmonary disease. *Lancet* 2015;385:899-909.

12. Boulet LP, Reddel H, Bateman E, et al. The Global Initiative for Asthma (GINA): 25 years later. *Eur Respir J* 2019; 54:1900598, doi:10.1183/13993003.00598-2019.

13. Huang K, Yang T, Xu J, et al. Prevalence, risk factors, and management of asthma in China: a national cross-sectional study. *Lancet* 2019; 394: 407-18.

14. Lebold KM, Jacoby DB, Drake MG. Inflammatory mechanisms linking maternal and childhood asthma. *J Leukoc Biol* 2020; 108:113-21.

15. Holgate ST. Innate and adaptive immune responses in asthma. *Nature medicine* 2012;18:673-83.

16. Holgate ST. Pathogenesis of asthma. *Clinical & Experimental Allergy* 2008;38:872-97.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

17. Cookson W. The alliance of genes and environment in asthma and allergy. *Nature* 1999;402:B5-11.

18. Kleeberger SR, Peden D. Gene-environment interactions in asthma and other respiratory diseases. *Annu Rev Med* 2005;56:383-400.

19. Matthew M, Denise F, Shaun H, et al. The global burden of asthma: executive summary of the GINA Dissemination Committee report. *Allergy* 2004;59:469-78.

20. Bobolea I, Arismendi E, Valero A, et al. Early Life Origins of Asthma: A Review of Potential Effectors. *J Investig Allergol Clin Immunol* 2019;29:168-79.

21. Haland G, Carlsen KC, Sandvik L, et al. Reduced lung function at birth and the risk of asthma at 10 years of age. *N Engl J Med* 2006;355:1682-9.

22. Fuchs O, Bahmer T, Rabe KF, et al. Asthma transition from childhood into adulthood. *The Lancet Respiratory Medicine* 2017;5:224-34.

23. Liu X, Olsen J, Agerbo E, et al. Maternal preeclampsia and childhood asthma in the offspring. *Pediatr Allergy Immunol* 2015;26:181-5.

24. Stokholm J, Sevelsted A, Anderson UD, et al. Preeclampsia Associates with Asthma, Allergy, and Eczema in Childhood. *Am J Respir Crit Care Med* 2017;195:614-21.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

25. Mirzakhani H, Carey VJ, McElrath TF, et al. Impact of Preeclampsia on the Relationship between Maternal Asthma and Offspring Asthma. An Observati on from the VDAART Clinical Trial. *Am J Respir Crit Care Med* 2019;199:32-42.

26. Byberg KK, Ogland B, Eide GE, et al. Birth after preeclamptic pregnancies: association with allergic sensitization and allergic rhinoconju nctivitis in late childhood; a historically matched cohort study. *BMC Pediatr* 2014;14:101.

27. Wilmink FA, den Dekker HT, de Jongste JC, et al. Maternal blood pressure and hypertensive disorders during pregnancy and childhood respiratory morbidi ty: the Generation R Study. *Eur Respir J* 2018;52.

28. Shaheen SO, Macdonald-Wallis C, Lawlor DA, et al. Hypertensive disorders of pregnancy, respiratory outcomes and atopy in childhood. *Eur Respir J* 2016;47:156-65.

29. Moher D, Liberati A, Tetzlaff J, et al. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *Ann Intern Med* 2009;151:264-9.

30. Stroup DF, Berlin JA, Morton SC, et al. Meta-analysis of observational studies in epidemiology: a proposal for reporting. Meta-analysis Of Observational Studies in Epidemiology (MOOSE) group. *Jama* 2000;283:2008-12.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

31. Li P, Xiong T, Hu Y. Hypertensive disorders of pregnancy and risk of asthma in offspring: protocol for a systematic review and meta-analysis. *BMJ open* 2020;10:e035145.

32. Andreas S. Critical evaluation of the Newcastle-Ottawa scale for the assessment of the quality of nonrandomized studies in meta-analyses. *European Journal of Epidemiology* 2010;25:603-5.

33. Nafstad P, Samuelsen SO, Irgens LM, et al. Pregnancy complications and the risk of asthma among Norwegians born between 1967 and 1993. *Eur J Epidemiol* 2003;18:755-61.

34. Rusconi F, Galassi C, Forastiere F, et al. Maternal complications and procedures in pregnancy and at birth and wheezing phenotypes in children. *Am J Respir Crit Care Med* 2007;175:16-21.

35. Aspberg S, Dahlquist G, Kahan T, et al. Confirmed association between neonatal phototherapy or neonatal icterus and risk of childhood asthma Is neonatal phototherapy associated with an increased risk for hospitalized childhood bronchial asthma. *Pediatr Allergy Immunol* 2010;21:e733-9.

36. Magnus MC, Haberg SE, Magnus P, et al. Pre-eclampsia and childhood asthma. *Eur*

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Respir J 2016;48:1622-30.

37. Duval S, Tweedie R. Trim and fill: a simple funnel-plot–based method of testing and adjusting for publication bias in meta-analysis. *Biometrics* 2000;56:455-63.

38. Wardenaar KJ, de Jonge P. Diagnostic heterogeneity in psychiatry: towards an empirical solution. *BMC medicine* 2013;11:1-3.

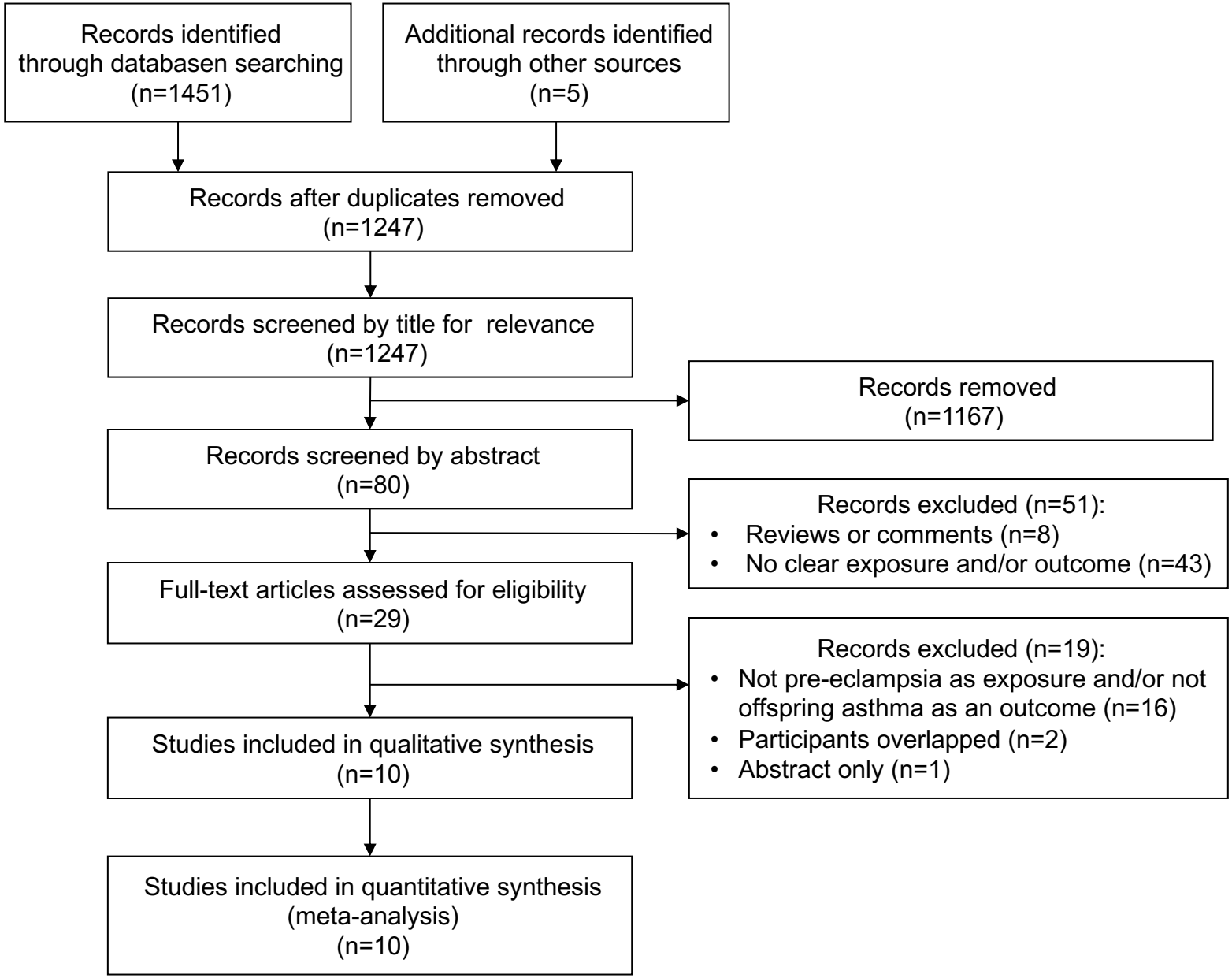
For peer review only

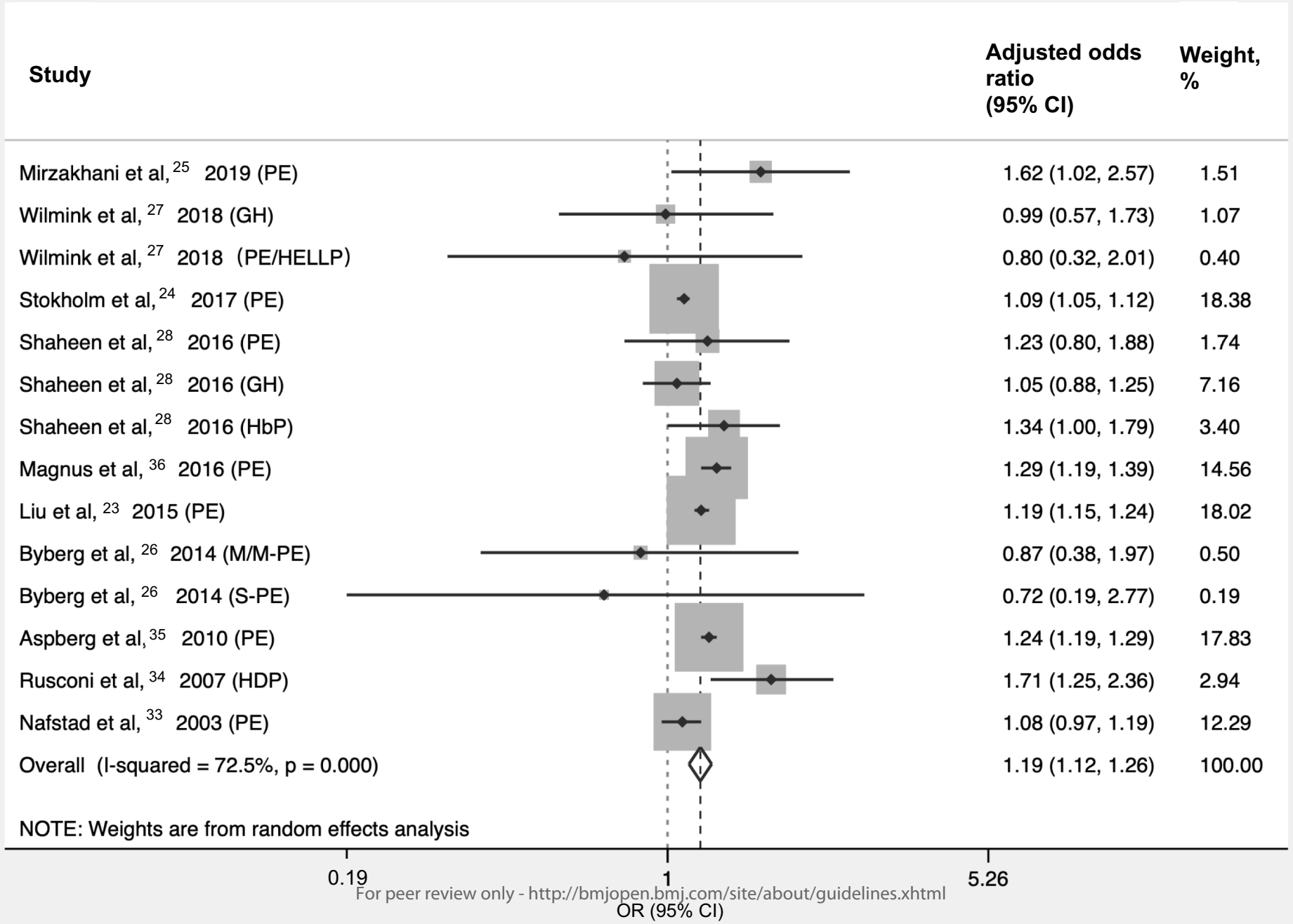
Figure 1. Flow diagram of the studies selected for inclusion in the systematic review.

Figure 2. Forest plot of studies investigating the association between hypertensive disorders in pregnancy (HDP) and asthma in offspring. Odds ratios (ORs) were calculated using a random-effects analysis. Diamonds indicate the effect size, size of markers, 95% CI. Abbreviations: PE, preeclampsia; GH, gestational hypertension; HELLP, hemolysis, elevated liver enzymes, and low platelets; HDP, hypertension disorders in pregnancy; HbP, hypertension before pregnancy; M/M-PE, mild/moderate preeclampsia; S-PE, severe preeclampsia; HDP, hypertension disorders in pregnancy; OR, odds ratio; CI, confidence intervals.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml


Records identified through databasen searching (n=1451)
Additional records identified through other sources (n=5)
Records after duplicates removed (n=1247)
Records screened by title for relevance (n=1247)
Records removed (n=1167)
Records screened by abstract (n=80)
Records excluded (n=51):
• Reviews or comments (n=8)
• No clear exposure and/or outcome (n=43)
Full-text articles assessed for eligibility (n=29)
Records excluded (n=19):
• Not pre-eclampsia as exposure and/or not offspring asthma as an outcome (n=16)
• Participants overlapped (n=2)
• Abstract only (n=1)
Studies included in qualitative synthesis (n=10)
Studies included in quantitative synthesis (meta-analysis) (n=10)



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplementary Online-Only Tables and Figures

Supplemental eTable 1. Search strategy for each electronic database.

Supplemental eTable 2 Newcastle-Ottawa Scale (NOS) for cohort studies.

Supplemental eTable 3. Effect estimates and adjusted confounders in the included studies.

Supplemental eTable 4. Sensitivity analysis of the included studies.

Supplemental eFigure 1. Forest plot of studies investigating the association between subtypes of HDP and asthma in offspring.

Supplemental eFigure 2. Funnel plot of published studies investigating the association between HDP and asthma in offspring.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplemental eTable 1. Search strategy for each electronic database.

#1	Search terms	No. of records returned
1	hypertensive disorders during pregnancy	7147
2	hypertensive disorders in pregnancy	7147
3	hypertensive disorders of pregnancy	7147
4	pregnancy-induced hypertension	2672
5	eclampsia	1665
6	pre-eclampsia	8158
7	preeclampsia	10799
8	preeclamptic pregnancy	464
9	chronic hypertension	3270
10	gestational hypertension	2538
11	chronic hypertension with superimposed preeclampsia	200
12	postpartum hypertension	988
13	gravid hypertension	65
14	maternal hypertensive disorders	50
15	(#1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14)	18801
16	asthma	53601
17	wheeze	9817
18	wheezing	9817
19	shortness of breath	5823
20	bronchial spasm	1749
21	bronchospasm	1749
22	bronchoconstriction	3413
23	bronchoconstrict	1505
24	bronchial hyperreactivity	5976
25	bronchial hyperresponsiveness	2321
26	reactive airway disease	280
27	obstructive lung disease	4734
28	airflow limitation	280
29	chronic obstructive respiratory disorder	7
30	chronic obstructive pulmonary disease	5110
31	(#16 OR #17 OR #18 OR #19 OR #20 OR #21 OR #22 OR #23 OR #24 OR #25 OR #26 OR #27 OR #28 OR #29 OR #30)	154781
32	(#15 AND #31)	1729
33	limit #32 to English language	1630
34	limit #33 to humans	1456

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplemental eTable 2. Newcastle-Ottawa Scale (NOS) for cohort studies.

Study	Item and scores								Scores and quality
	Representiveness of the exposed cohort（1）	Selection of the non-exposed cohort（1）	Ascertainment of exposure（1）	Demonstration that the outcome of interest was not present at the start of the study（1）	Comparability of cohorts based on the design or analysis（2）	Assessment of outcome（1）	Was follow up long enough for outcomes to occur（1）	Adequacy of follow up of cohorts（1）	
Mirzakhani et al (2019)	0	1	1	1	2	0	0	1	6 (M)
Wilmink et al (2018)	1	1	1	1	1	0	1	1	7 (H)
Stokholm et al (2017)	1	1	1	1	2	1	1	1	9 (H)
Shaheen et al (2016)	1	1	1	1	2	0	1	1	8 (H)
Magnus et al (2016)	1	1	1	1	2	1	1	1	9 (H)
Liu et al (2015)	1	1	1	1	2	1	1	1	9 (H)
Byberg et al (2014)	0	1	1	1	2	0	1	1	7(H)
Aspberg et al (2010)	1	1	1	1	0	1	1	0	6 (M)
Rusconi et al (2007)	1	1	1	1	1	0	1	1	7(H)
Nafstad et al (2003)	1	1	1	1	0	1	1	1	7(H)

Note: The quality of the studies was divided into the following 3 categories: high quality (scores 7–9), moderate quality (scores 4–6) and low quality (scores 0–3).

Abbreviations: M, moderate quality; H, high quality.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplemental eTable 3. Effect estimates and adjusted confounders in the included studies.

Author (year)	Effect estimate	Confounders adjusted					
		Mothers				Offspring	
		Maternal asthma	Maternal smoking during pregnancy	Maternal BMI	Others	Birthweight and gestational age	Others
Mirzakhani et al (2019)	adjusted RR [a]: asthma: 1.62 (1.02–2.57)	/	/	maternal BMI at first appointment	race, maternal age, trial arm, maternal baseline vitamin D, clinical center, maternal marital status, maternal education, gestational diabetes, mode of delivery, parity	gestational age, birthweight	sex
	adjusted HR [b]: asthma: 2.68 (1.30–5.61) asthma and recurrent wheeze: 4.73 (2.20–10.07)	yes	/	maternal BMI at first appointment	race, maternal age, trial arm, maternal baseline vitamin D, clinical center, maternal marital status, maternal education, gestational diabetes, mode of delivery, parity	gestational age, birthweight	sex
Wilmink et al (2018)	adjusted OR: current asthma [c]: 0.99 (0.57, 1.73) (GH); 0.80 (0.32, 2.01) (PE/HELLP) current wheezing [d]: 0.95 (0.63, 1.42) (GH); 0.70 (0.36, 1.39) (PE/HELLP)	/	yes	pre-pregnancy BMI	maternal age, maternal ethnicity, educational level, nulliparity, folic acid use	/	sex

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplemental eTable 3. Effect estimates and adjusted confounders in the included studies (continued).

Author (year)	Effect estimate	Confounders adjusted					
		Mothers				Offspring	
		Maternal asthma	Maternal smoking during pregnancy	Maternal BMI	Others	Birthweight and gestational age	Others
Stokholm et al (2017)	adjusted IRR: Registry-based cohort: 1.09 (1.05, 1.12)	/	/	/	mode of delivery, parity, multiple births, season of birth, maternal age, and mother's disease [e]	birth weight, gestational age	sex, age and calendar year, and season of birth
	adjusted OR [f]: COPSAC$_{2000}$ Birth Cohort: 1.13 (0.31, 4.13)	yes	/	/	type of delivery	/	/
Shaheen et al (2016)	adjusted OR: 1.20 (0.78, 1.83) (preeclampsia) 1.04 (0.87, 1.24) (GH) 1.33 (1.00, 1.78) (HbP)	yes	/	pre-pregnancy BMI	maternal age, maternal parity, maternal education, multiple pregnancy and maternal ethnicity	/	sex
	adjusted OR: 1.23 (0.80, 1.88) (preeclampsia) 1.05 (0.88, 1.25) (GH) 1.34 (1.00, 1.79) (HbP)	yes	yes	pre-pregnancy BMI	maternal age, maternal parity, maternal education, multiple pregnancy and maternal ethnicity	/	sex

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplemental eTable 3. Effect estimates and adjusted confounders in the included studies (continued).

Author (year)	Effect estimate	Confounders adjusted					
		Mothers				Offspring	
		Maternal asthma	Maternal smoking during pregnancy	Maternal BMI	Others	Birthweight and gestational age	Others
Magnus et al (2016)	adjusted RR: registry-based study: 1.31 (1.22, 1.41)	yes	/	/	maternal age, parity, education	/	sex
	adjusted RR: registry-based study: 1.29 (1.19–1.39)	yes	yes	/	maternal age, parity, education	/	sex
	adjusted OR: sibling pair analysis: 1.07 (0.87, 1.33)	yes	/	/	maternal age, parity, education	/	sex
	adjusted RR: MoBa study: 1.19 (0.99–1.44)	yes	yes	pre-pregnancy BMI	maternal age, parity, education	/	sex
Liu et al (2015)	adjusted IRR: nested case-control study: 1.19 (1.15, 1.24)	yes	yes	/	maternal age, maternal social status at birth, parity, and maternal country of origin	/	/
	adjusted IRR: case-sibling study: 1.06 (0.98, 1.14)	/	yes	/	maternal age, maternal social status at birth, and parity	/	sex, calendar year of birth

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplemental eTable 3. Effect estimates and adjusted confounders in the included studies (continued).

Author (year)	Effect estimate	Confounders adjusted					
		Mothers				Offspring	
		Maternal asthma	Maternal smoking during pregnancy	Maternal BMI	Others	Birthweight and gestational age	Others
Byberg et al (2014)	adjusted OR: FU1, asthma: 0.87 (0.38, 1.97) (mild/moderate preeclampsia) 0.72 (0.19, 2.77) (severe preeclampsia)	yes	yes	maternal BMI	cesarean section, maternal age	[g]	sex, being firstborn
	adjusted OR: FU2, current asthma [h]: 1.11 (0.45, 2.73) (mild/moderate preeclampsia) 0.69 (0.15, 3.18) (severe preeclampsia)	yes	yes	maternal BMI	cesarean section, maternal age	[g]	sex, being firstborn
Aspberg et al (2010)	adjusted OR: 1.24 (1.19–1.29)	/	yes	/	maternal age, parity, and involuntary childlessness	/	year of birth
Rusconi et al (2007)	adjusted OR: transient early wheezing [i]: 1.47 (1.14–1.89) persistent wheezing [j]: 1.71 (1.25–2.36) late-onset wheezing [k]: 1.45 (1.06–1.48)	/	yes	/	study center, parental atopy, maternal age, socioeconomic status, season of the questionnaire's completion, and person who completed the questionnaire	/	sex, low birth weight, siblings

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplemental eTable 3. Effect estimates and adjusted confounders in the included studies (continued).

Author (year)	Effect estimate	Confounders adjusted					
		Mothers				Offspring	
		Maternal asthma	Maternal smoking during pregnancy	Maternal BMI	Others	Birthweight and gestational age	Others
Nafstad et al[l] (2003)	adjusted OR: 1.08 (0.97–1.19) (total)	/	/	/	hyperemesis, maternal age, maternal education, plurality, single parent family, death or emigration during the study period, maternal atopy, diabetes mellitus, lower respiratory tract infections, and other maternal infections during pregnancy, other pregnancy complications [m]	/	sex, birth order, year of birth, geographical district of birth

Note:

[a] Mothers with preeclampsia compared with mothers without preeclampsia;

[b] Mothers with asthma with preeclampsia compared with mothers without asthma without preeclampsia;

[c] Current asthma was defined as previous diagnosis of asthma by physician with either current wheezing or the use of inhaled medication in the past 12 months;

[d] Current wheezing, physician diagnosed asthma and use of inhaled medication (bronchodilators or corticosteroids) in the past 12 months was assessed by a parental questionnaire at the age of 10 years;

[e] Mother's disease, not mentioned;

[f] Mothers with asthma with preeclampsia compared with mothers with asthma without preeclampsia (all mothers were diagnosed with asthma in the whole cohort);

[g] All children were born full term;

[h] Current asthma at FU2 was defined as asthma ever in addition to asthma symptoms or the use of asthma medication during the last 12 months;

[i] Transient early wheezers: those who had at least one lower respiratory tract illness with wheezing (bronchiolitis, asthmatic bronchitis, asthma or wheezing) in the first 2 yr of life and no wheezing in the last 12 mo;

[j] Persistent wheezers: those who had at least one lower respiratory tract illness with wheezing in the first 2 yr of life and wheezing in the last 12 mo;

[k] Late-onset wheezers: those with no lower respiratory tract illness with wheezing in the first 2 yr of life but who had wheezing in the last 12 mo (late-onset wheezers).

[l] Only persons with severe asthma are registered in NIAR;

[m] Pregnancy complications, including hemorrhage, anemia, cervix insufficiency, placenta dysfunction, rhesus immunization, and preterm contractions.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Abbreviations: HDP, hypertension disorders in pregnancy; GH, gestational hypertension; HELLP, hemolysis, elevated liver enzymes, and low platelets; HbP, hypertension before pregnancy; VDRRT, Vitamin D Antenatal Asthma Reduction Trial; ISSHP, International Society for the Study of Hypertension in Pregnancy; ACOG, American College of Obstetricians and Gynaecologists; $COPSAC_{2000}$, Copenhagen Prospective Studies on Asthma in $Childhood_{2000;}$ ALSPAC, Avon Longitudinal Study of Parents and Children; DMBR, Danish Medical Birth Registry; ISAAC, International Study of Asthma and Allergies in Childhood; SIDRIA, Italian Studies of Respiratory Disorders in Childhood and the Environment; MBRN, Medical Birth Registry of Norway; NIAR, National Insurance Administration Register.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplemental eTable 4. Sensitivity analysis of the included studies.

Study omitted	Estimate	95% CI
Mirzakhani et al, 2019 (PE)	1.18	1.11 - 1.25
Wilmink et al, 2018 (GH)	1.19	1.12 - 1.26
Wilmink et al, 2018 （PE/HELLP）	1.19	1.12 - 1.26
Stokholm et al, 2017 (PE)	1.21	1.15 - 1.27
Shaheen et al, 2016 (PE)	1.18	1.11 - 1.25
Shaheen et al, 2016 (GH)	1.20	1.12 - 1.26
Shaheen et al, 2016 (HbP)	1.18	1.11 - 1.23
Magnus et al, 2016 (PE)	1.16	1.09 - 1.23
Liu et al, 2015 (PE)	1.18	1.09 - 1.27
Byberg et al, 2014 (M/M-PE)	1.18	1.11 - 1.25
Byberg et al, 2014 (S-PE)	1.18	1.11 - 1.25
Aspberg et al, 2010 (PE)	1.17	1.10 - 1.24
Rusconi et al, 2007 (HDP)	1.17	1.11 - 1.24
Nafstad et al, 2003 (PE)	1.19	1.11 - 1.26

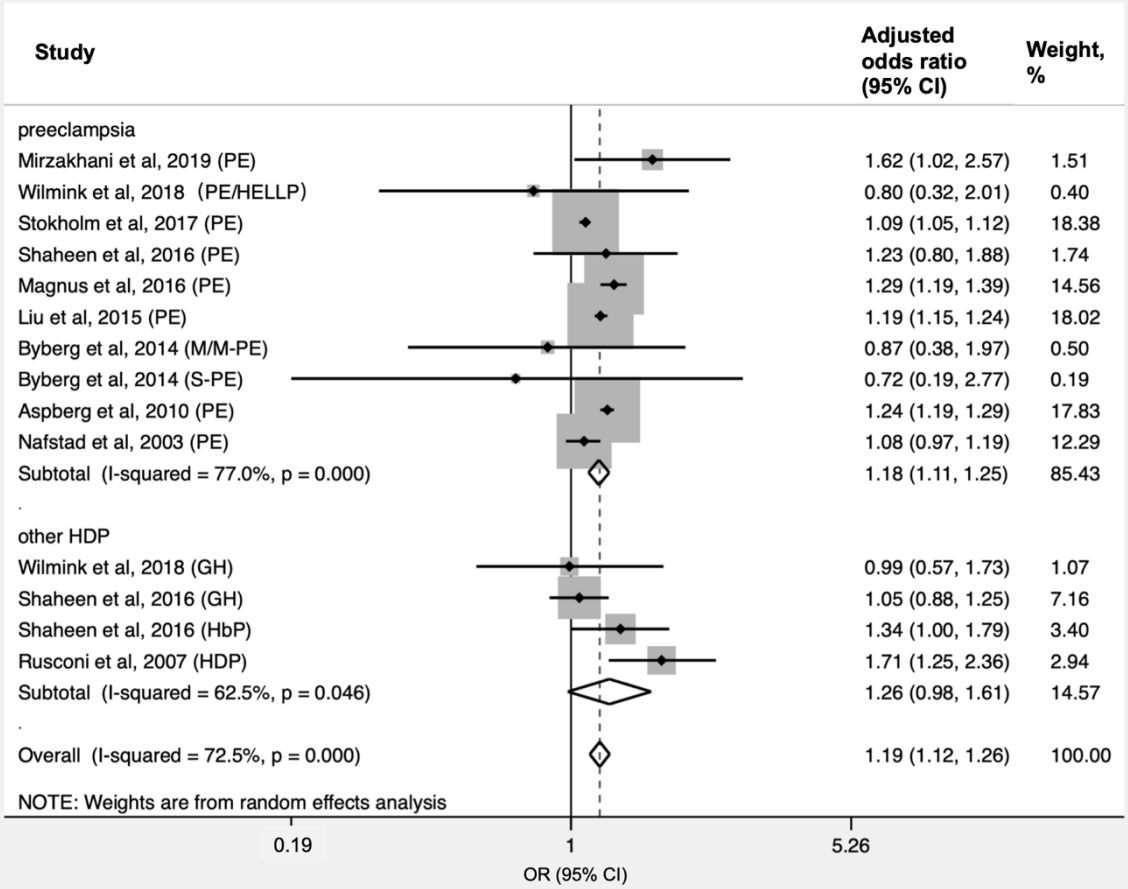
The estimate in each row indicates the pooled ORs after excluding that particular study.

Abbreviations: HDP, hypertension disorders in pregnancy; GH, gestational hypertension; HELLP, hemolysis, elevated liver enzymes, and low platelets; HbP, hypertension before pregnancy; CI, confidence intervals.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplemental eFigure 1. Forest plot of studies investigating the association between subtypes of HDP and asthma in offspring.

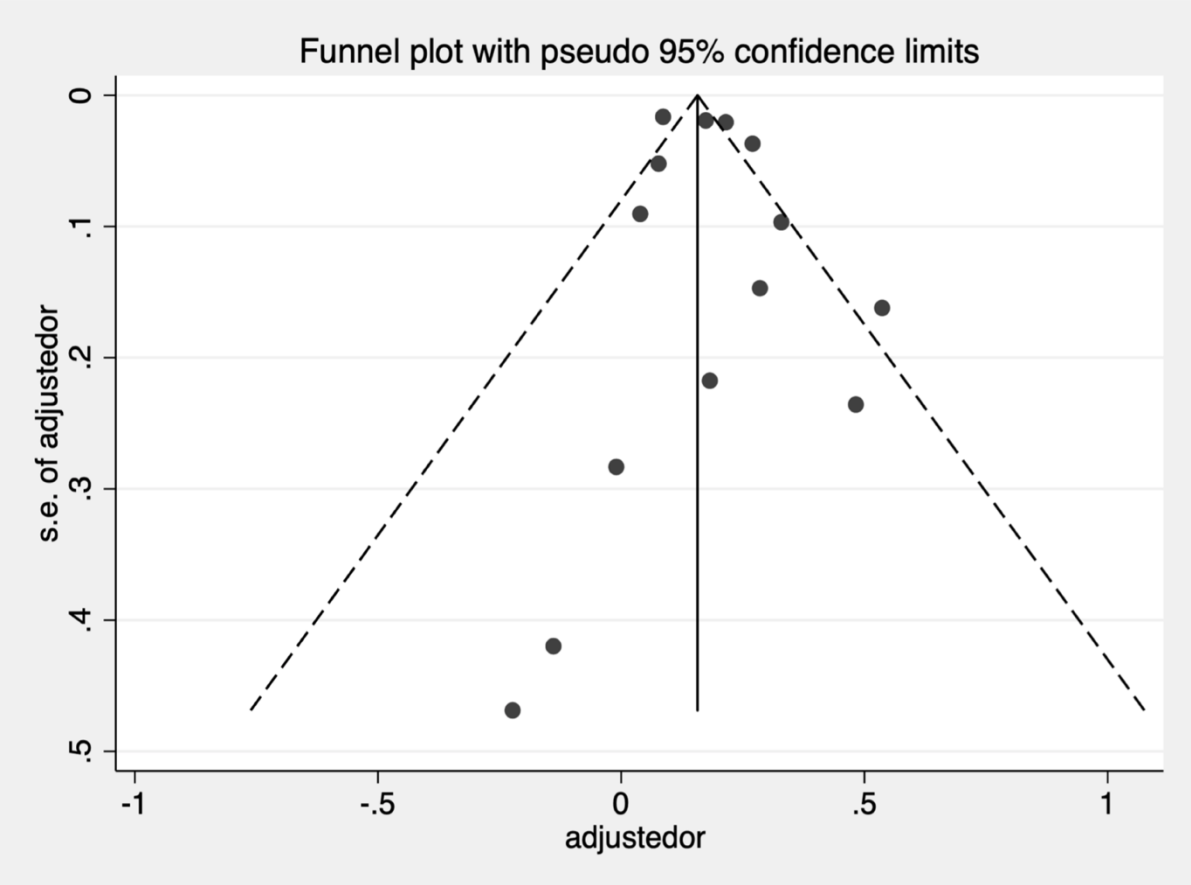


Forest plots displaying adjusted estimates examining the association between subtypes of HDP and asthma in offspring.

Abbreviations: PE, preeclampsia; GH, gestational hypertension; HELLP, hemolysis, elevated liver enzymes, and low platelets; HDP, hypertension disorders in pregnancy; HbP, hypertension before pregnancy; M/M-PE, mild/moderate preeclampsia; S-PE, severe preeclampsia; HDP, hypertension disorders in pregnancy; OR, odds ratio; CI, confidence intervals.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Supplemental eFigure 2. Funnel plot of published studies investigating the



association between HDP and asthma in offspring.

Abbreviations: ORs, odds ratios

For peer review only

MOOSE Checklist for Meta-analyses of Observational Studies

Item No	Recommendation	Reported on Page No
Reporting of background should include		
1	Problem definition	5
2	Hypothesis statement	6
3	Description of study outcome(s)	6,7
4	Type of exposure or intervention used	5
5	Type of study designs used	7
6	Study population	7
Reporting of search strategy should include		
7	Qualifications of searchers (eg, librarians and investigators)	None
8	Search strategy, including time period included in the synthesis and key words	7,8
9	Effort to include all available studies, including contact with authors	8
10	Databases and registries searched	7
11	Search software used, name and version, including special features used (eg, explosion)	8
12	Use of hand searching (eg, reference lists of obtained articles)	8
13	List of citations located and those excluded, including justification	8
14	Method of addressing articles published in languages other than English	8
15	Method of handling abstracts and unpublished studies	8
16	Description of any contact with authors	8
Reporting of methods should include		
17	Description of relevance or appropriateness of studies assembled for assessing the hypothesis to be tested	8
18	Rationale for the selection and coding of data (eg, sound clinical principles or convenience)	8,9
19	Documentation of how data were classified and coded (eg, multiple raters, blinding and interrater reliability)	9
20	Assessment of confounding (eg, comparability of cases and controls in studies where appropriate)	9,11
21	Assessment of study quality, including blinding of quality assessors, stratification or regression on possible predictors of study results	10
22	Assessment of heterogeneity	11
23	Description of statistical methods (eg, complete description of fixed or random effects models, justification of whether the chosen models account for predictors of study results, dose-response models, or cumulative meta-analysis) in sufficient detail to be replicated	10,11
24	Provision of appropriate tables and graphics	8,11
Reporting of results should include		
25	Graphic summarizing individual study estimates and overall estimate	16
26	Table giving descriptive information for each study included	12,15
27	Results of sensitivity testing (eg, subgroup analysis)	17~19
28	Indication of statistical uncertainty of findings	None

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Item No	Recommendation	Reported on Page No
Reporting of discussion should include		
29	Quantitative assessment of bias (eg, publication bias)	22
30	Justification for exclusion (eg, exclusion of non-English language citations)	22
31	Assessment of quality of included studies	21
Reporting of conclusions should include		
32	Consideration of alternative explanations for observed results	24
33	Generalization of the conclusions (ie, appropriate for the data presented and within the domain of the literature review)	24
34	Guidelines for future research	24
35	Disclosure of funding source	24

From: Stroup DF, Berlin JA, Morton SC, et al, for the Meta-analysis Of Observational Studies in Epidemiology (MOOSE) Group. Meta-analysis of Observational Studies in Epidemiology. A Proposal for Reporting. *JAMA*. 2000;283(15):2008-2012. doi: 10.1001/jama.283.15.2008.

Transcribed from the original paper within the NEUROSURGERY® Editorial Office, Atlanta, GA, United Sates. August 2012.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml